

Industry Level Signals from Institutional Portfolio Reallocations

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Abstract

This paper develops a framework to extract and quantify the information embedded in institutional investors' industry-level demand. Changes in portfolio allocations across industries are interpreted as signals about future returns, and investors are identified as industry experts based on the historical accuracy of their allocation shifts in predicting subsequent industry performance. Aggregating these expert signals yields a novel measure, Revealed Industry Information (RII). I show that RII strongly predicts future industry returns and generates economically significant abnormal performance that cannot be explained by standard asset pricing factors or industry characteristics. A long-short portfolio formed on RII earns a six-factor alpha of 1.6% per quarter after controlling for the Fama-French five factors and momentum.

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1 Introduction

Within 45 days after the end of each quarter, institutional investors disclose their portfolio holdings, providing a glimpse of "who owns what". Analogous to consumer theory, in which consumption bundles reveal preferences, portfolio choices in financial markets reveal investors' expectations. Since expectations are shaped by information, these portfolio choices serve as revealed signals about future prices. Do these revealed signals contain information orthogonal to the known set of price-relevant information (e.g., characteristics)?

Theoretically yes. In a competitive equilibrium with asymmetric information, where acquiring information is costly, prices only partially reflect all available information; otherwise, investors would lack incentives to acquire or process it (Grossman and Stiglitz (1980)). Therefore, even in a market with fully rational agents, revealed portfolio choices may carry information orthogonal to publicly available information set.

Empirically, however, this question remains largely unexplored. This paper aims to extract information from investors' holdings and test whether it provides insights orthogonal to the existing set of price-relevant information. I interpret changes in investor demand as signals about future asset prices and assess the accuracy of these signals based on their historical co-movement with realized returns. I then aggregate these signals to construct an industry-level measure of expected returns and test its predictive power. My primary results indicate that this measure significantly predicts future industry prices- which suggest that investors observed holdings may contain some hidden information on future states of the world.

This analysis follows several steps. The first step is to define the universe of assets and investors in a way that minimizes the signal-to-noise ratio. The investor universe is constructed from 13F filings, which report the quarterly holdings of institutional investors, while households are aggregated as a residual category. Instead of focusing on individual assets, I examine industries for several reasons. First, industries provide a natural unit of information because industry price movements are likely driven by key state variables

such as oil prices, interest rates, and technological shifts. Second, investors specialize in industries due to economies of scale in learning (Van Nieuwerburgh and Veldkamp (2005)). Third, aggregating at the industry level reduces idiosyncratic noise, improving signal quality. Finally, industry return movements remain an open question in asset pricing, as their true factor premia are unknown and industry loadings on common factors are unstable over time (Fama and French (1997)).

The second step is to refine the reported holdings into a clearer signal. Portfolio holdings are typically disclosed in dollar values. However, this raw signal is difficult to interpret in an economically meaningful way. To extract a more relevant signal, I focus on portfolio weights instead.¹ Portfolio weights capture investors' expectations about the distribution of future returns, incorporating not only fundamental signals but also behavioral biases, hedging motives, and risk management strategies.

The challenge is to disentangle the price-driven component of demand from other demand motives, such as hedging, liquidity needs, or behavioral biases. To isolate the price-relevant component, I examine historical co-movements between investor demand and realized future returns. Specifically, I perform time-series regressions of realized returns on historical investor demand while controlling for known characteristics. This approach explicitly identifies investors whose demand contains price-relevant signals that are orthogonal to existing asset characteristics. Approximately two-thirds of institutional investors were identified as experts (those with accurate signals) in at least one industry, with the majority being investment advisors, mutual funds, and investment banks—key players expected to possess industry-specific information. Interestingly, while they hold on average 40 industries in their portfolios, they are identified as experts in only four on average, highlighting the specialization and heterogeneity of institutions across industries.

Once the price-informative signals are identified, the next challenge is determining how

¹Two other alternatives are deviations from a fully diversified portfolio or tilts. I have done all empirical tests for both weights and tilts. They yield similar results. Another approach is the market share of each investor in an industry, a method suggested by my friend from UIC. I have not derived market share empirical results yet.

to aggregate them into an industry-level measure of information. I construct Revealed Industry Information (RII) by averaging the signal across investors who have been identified as experts. Specifically, at each quarter, RII assigns each industry a score, calculated as the mean change in that industry’s experts demand. This equal-weighted approach assumes that all experts’ signals are equally reliable. To refine this measure, I introduce an accuracy adjustment, where each investor’s contribution is weighted by their historical predictive accuracy, captured by their coefficient in time-series regressions. This ensures that signals from investors with a stronger track record of forecasting industry returns are given greater influence in the final measure. ²

The final step is to test whether the revealed industry information extracted from investors’ portfolio choices contains new information about future industry prices. To assess this, I perform a Fama-MacBeth regression across 48 industries from 1980 to 2017, controlling for well-known asset pricing factors such as the Fama-French five factors and momentum. Preliminary results indicate that RII significantly predicts future returns. A one standard deviation increase in RII corresponds to a 76 basis points increase in next-period industry returns, highlighting the strong predictive power of institutional demand signals.

Next, I construct five portfolios by sorting industries based on RII. Industries in the top quintile consistently outperform those in the bottom quintile by approximately 2% per quarter (8% annually). To further evaluate the robustness of this predictive signal, I examine whether these portfolio returns can be explained by standard asset pricing factors. The results show that the long-short portfolio earns a six-factor alpha of 1.6 percent per quarter, after controlling for the Fama-French five factors and momentum.

This paper contributes to the literature that uses portfolio holdings to infer investor information and skill. Prior studies show that holdings contain valuable information about future returns and manager ability. Grinblatt and Titman (1993) and Daniel, Grinblatt, Titman,

²I plan to explore a Bayesian approach, such as Bayesian Model Averaging (BMA) or a hierarchical Bayesian model, which assigns dynamic weights to experts based on their historical predictive accuracy. Unlike a simple average, these methods prioritize investors whose tilts have consistently aligned with realized returns, potentially yielding a more precise estimate of industry expected returns.

and Wermers (1997) develop holdings-based measures of performance, while Cohen, Coval, and Pastor (2005) and Wermers, Yao, and Zhao (2012) extract information from portfolio holdings to identify skilled investors and predict stock returns. Similarly, Cohen, Polk, and Silli (2009) show that managers' highest-conviction positions outperform, suggesting that holdings reveal information not fully reflected in prices. This paper differs from these studies in two important ways. First, rather than evaluating investor skill at the portfolio or stock level, I identify expertise at the industry level. Second, instead of using holdings to evaluate managers, I aggregate information embedded in investors' demand to construct a new industry-level measure, Revealed Industry Information (RII), and study its implications for expected returns. This paper also contributes to the literature on industry specialization and industry return predictability. Kacperczyk, Sialm, and Zheng (2005) show that mutual funds with more concentrated industry portfolios outperform, while Van Nieuwerburgh and Veldkamp (2005) argue that specialization naturally arises because investors face increasing returns to acquiring information. In the asset-pricing literature, Fama and French (1997) highlight the difficulty of explaining industry returns using standard risk factors, and Moskowitz and Grinblatt (1999) document that momentum forecasts industry returns. More recently, studies such as Rapach, Strauss, Tu, and Zhou (2015) use statistical and machine-learning methods to forecast industry returns. In contrast, I propose a new source of predictive information: the revealed demand of investors identified as industry experts. The results suggest that industry return predictability is partly driven by information embedded in investors' portfolio reallocations, which contains information beyond standard industry characteristics and asset-pricing factors.

The remainder of the paper is structured as follows: Section 2 describes the data. Section 3 elaborates on empirical framework and describe the results. Finally, Section 4 concludes.

2 Data Description

Before delving into quantifying information, I summarize the data and describe holdings. This section is structured into 4 sections. Section 2.1 provides an overview of stock characteristics data. Section 2.2 discusses stock-level institutional holdings. Section 2.3 elaborates on the aggregation of characteristics and holdings to form industry level characteristics and holdings. Finally, section 2.4 introduces tilt, as the variable which represents demand, along with portfolio weights- and summarizes its dynamics.

2.1 Stock Characteristics

The data on stock prices, dividends, returns, and shares outstanding are sourced from the Center for Research in Security Prices (CRSP) Monthly Stock Database. I restrict my sample to ordinary common shares (i.e., share codes 10, 11, 12, and 18) traded on the New York Stock Exchange (NYSE), the American Stock Exchange, and Nasdaq (i.e., exchange codes 1, 2, and 3). Additionally, I restrict the sample to stocks with non-missing prices and shares outstanding. Accounting data are sourced from the Compustat North America Fundamentals Annual and Quarterly Databases. I merge CRSP data with the most recent Compustat data available at least 6 months and no more than 18 months prior to the trading date. The 6-month lag ensures that the accounting data were public at the time of trading.

In addition to market equity, the characteristics include profitability, investment, dividends-to-book equity, and market beta. The selection of book equity, profitability, and investment is motivated by the Fama-French five-factor model, which describes the cross-section of stock returns. Dividends and market beta have a long tradition in empirical asset pricing as proxies for fundamentals and systematic risk, respectively.

The construction of these characteristics follows Fama and French (2015), summarized as follows: profitability is the ratio of operating profits to book equity; investment is the annual log growth rate of assets; dividends-to-book equity is the ratio of annual dividends per

split-adjusted share times shares outstanding to book equity. Market beta is estimated by regressing monthly excess returns over the 1-month Treasury-bill rate onto excess market returns using a 60-month moving window (requiring at least 24 months of non-missing returns). I winsorize profitability, investment, and market beta at the 2.5th and 97.5th percentiles, while dividends-to-book equity, being positive, is winsorized at the 97.5th percentile.

Following Fama and French (1993), I focus on ordinary common shares that are not foreign or real estate investment trusts (REITs) (i.e., share codes 10 and 11) with non-missing characteristics and returns. These stocks form the investment universe, while foreign stocks (share code 12), REITs (share code 18), or those with missing characteristics or returns are excluded.

2.2 Institutional Stock Holdings

The data on institutional stock holdings are from the Thomson Reuters Institutional Holdings Database (s34 file), derived from SEC Form 13F filings. Institutional managers with over \$100 million in Section 13(f) securities are required to file this form, which includes long positions but excludes short positions, cash, and bonds.

Institutions are classified into six types: banks, insurance companies, investment advisors, mutual funds, pension funds, and other 13F institutions. Investment advisors include hedge funds unless separately categorized as mutual funds. The "other" group includes endowments, foundations, and non-financial corporations.

I merge institutional holdings data with CRSP-Compustat data using CUSIP numbers, excluding unmatched holdings. I compute the dollar holdings for each stock as the product of price and shares held, and the total assets under management (AUM) as the sum of dollar holdings for each institution. Portfolio weights are calculated as the ratio of dollar holdings to AUM.

Market clearing requires that shares outstanding equal the sum of shares held across all investors. For each stock, I calculate the household sector holdings as the difference between

shares outstanding and the total shares held by 13F institutions. The household sector includes direct household holdings and smaller institutions not required to file Form 13F, as well as institutions with less than \$10 million in AUM or no stocks in the investment universe.

2.3 Industry Level Data

The asset level dataset comprises approximately 80 million asset-institution-quarter observations from 1980, covering roughly 8,000 unique institutions over the entire period. Now, I aggregate asset-level data into industry-level data. First, I classify assets into 48 industries using the Fama-French (F-F) classification. Firms that do not fall into any of these categories are assigned to a 49th industry, labeled "NS" (non-specified).

Then, for each industry, characteristics are constructed as follows. Book equity (Be) and market equity (Me) for an industry are obtained by summing the book equity and market equity of all assets within that industry. Growth in assets (Gat), profitability, dividends-to-book equity (Div/Be), and industry returns are computed as the market-value weighted average of these characteristics across all assets in the industry, with weights determined by each asset's market equity.

Finally, industry-level holdings are computed similarly: the holding of investor i at time t in industry j is the sum of the dollar value of that investor's holdings across all assets in the industry. After this aggregation, with 49 industries, 152 quarters, and 7,966 investors, the dataset shrinks to approximately 7.8 million observations.

Previous analysis³ has shown that investors significantly deviate from a fully diversified portfolio at the asset level. Here, I extend this investigation to measure diversification at the industry level—that is, how investors allocate capital across different industries. To assess this, Table 1 presents the average number of industries in investors' portfolios. This measure is constructed as follows. For each investor at each point in time, I compute the number of

³My last draft

industries they are invested in. I then drive statistics across investors at each point in time and then take the average of statistics over time.

On average, 1,704 institutions were active per quarter. The number of industries in an investor’s portfolio ranges from 1 (fully concentrated) to 49 (fully diversified). 25% of investors hold positions in 21 or fewer industries, while the mean number of industries per portfolio is 29, which is close to half of all available industries. The median is also 29, suggesting that a typical institutional investor is far from fully diversified across industries. The main takeaway is that even at the industry level, where the asset universe is significantly smaller, investors remain far from holding fully diversified portfolios. This provides evidence that some investors may have specialized expertise in specific industries.

	Count	Mean	Std Dev	Min	10th	25th	Median	75th	90th	Max
# of industries	1704	29	11	1	14	21	29	38	45	49

Table 1: Summary statistics of the number of industries in investors’ portfolios. At each point in time, I determine the number of industries each investor is invested in and compute the corresponding summary statistics. I then take the average across time and report the results.

Figure 1 examines whether diversification has changed over time. Rather than averaging these statistics over time-what I did in the table 1- I plot their time series to observe their temporal dynamics. Two notable patterns emerge. First, the 25th percentile has declined over time, indicating that an increasing number of investors are concentrating their holdings in fewer industries. Second, the standard deviation has increased over time. This rise is driven by the combination of a stable 75th percentile and a declining 25th percentile, leading to greater dispersion in industry allocations across investors.

Figure 2 supports this poor diversification from a more precise angle. It presents the distribution of the number of industries held by investors across investor-quarter observations. The dataset contains approximately 260,000 investor-quarter observations. For each investor-quarter, I compute the number of industries an investor holds and plot the distribu-

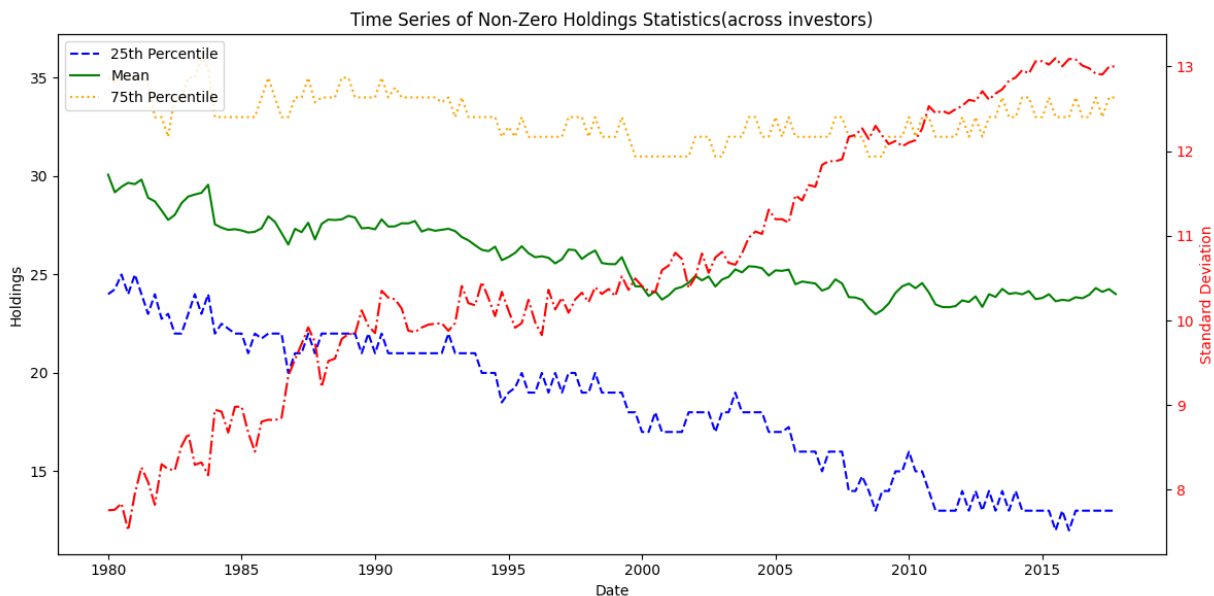


Figure 1: This graph drives number of industries in portfolios for each investor and then plots some of its statistics over time.

tion in the top panel. The horizontal axis represents the number of industries held (ranging from 1 to 49), while the vertical axis shows the frequency of observations.

Several notable patterns emerge. First, the distribution is approximately normal, with a large proportion of investors partially diversified, holding between 25 and 30 industries in their portfolios. Why this graph appears to be normally distributed around 25–30 industries is an interesting question. Could it be related to the social structure of information

Second, a greater number of investors appear in the right tail of the distribution. The bottom panel presents the same distribution but aggregated into bins of five to make the numbers easier to read. It highlights that 5 percent of investors hold fewer than five industries, while approximately 19 percent hold fewer than 15 industries in their portfolios. This suggests that while most investors exhibit moderate diversification, a non-negligible fraction prefers highly concentrated industry exposures.

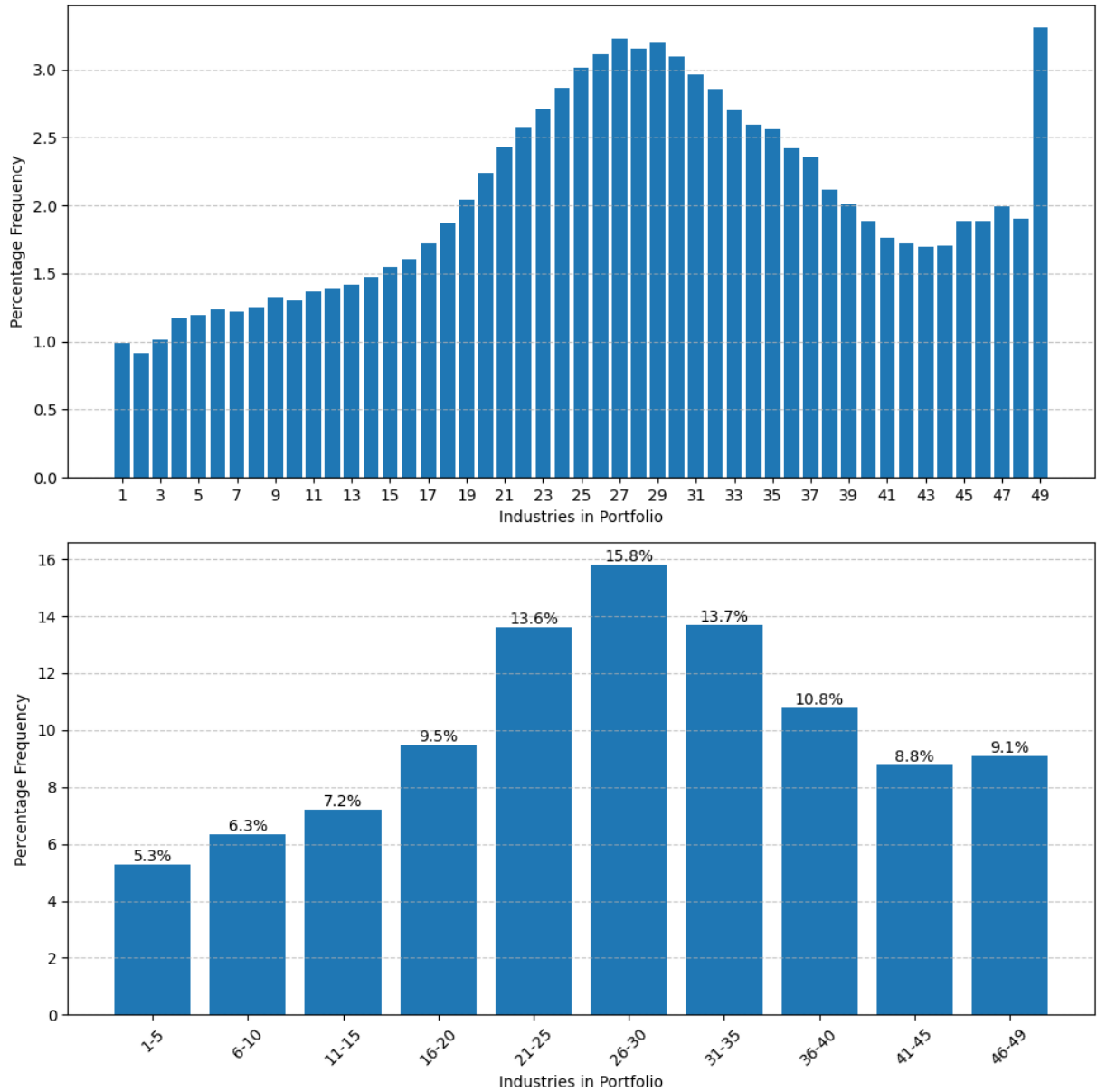


Figure 2: This figure is intended to show distribution of number of industries in portfolios. Each investor-quarter is treated as one observation. Then, each investor at each quarter holds a certain number between 1 to 49 in her portfolios. The top panel plots this distribution for all possible number of industries. The bottom one report the same but bins then in 5 industries to make numbers more readable.

2.4 Tilt and its Dynamics

This section aims to describe tilt and its dynamics. Tilt is one of the variables which represents demand in this study. The other one is portfolio weights which have the same statistical properties. This section elaborates on some statistics of tilts, and demonstrates that tilt varies significantly across three dimensions: over time, across investors, and across industries.

Poor industry diversification, demonstrated in the last sections, suggests that investors may operate within a narrow investment universe. Therefore, this causes a plethora of zero holdings. How to handle this zero holdings is still an ongoing issue in literature. One simple way suggested by literature is to define an investment universe based on the historical holdings.

To address this, I follow the literature and extend the asset-level approach to the industry level, defining an investor's investment universe as the set of all industries they have invested in over the past 11 quarters. If an industry is in an investor's universe but is currently absent in her portfolios, it is interpreted as a low expected return from that investor. If investor has zero holding on an asset which is out of her investment universe, that does not translate in a low expected return of that investor in that industry. I label investment universe of investor i at quarter t as $U_{i,t}$.

To construct each investor's universe, I first exclude investors with fewer than 30 quarters. These short-tenure institutions are aggregated into a single "stacked" investor, preserving their collective impact while maintaining dataset consistency. For each remaining investor, I define their investment universe dynamically based on their past holdings. Specifically, an investor's universe in its 12th quarter consists of all industries it invested in during the first 11 quarters. From the 12th quarter onward, an industry remains in the investor's universe if they currently hold it or have held it at any point in the past 11 quarters. This approach provides a realistic and adaptive measure of each investor's accessible industries, ensuring that historical investment behavior informs current industry selection. This reduces the

dataset to 7.4 million observations and 3,107 unique investors.

Then, to measure how an investor’s portfolio deviates from the market portfolio at the industry level, I define *tilt* as:

$$\text{tilt}_{ijt} = \begin{cases} \frac{\text{Holding}_{ijt}}{\text{AUM}_{it}} - \frac{\text{ME}_{jt}}{\sum_j \text{ME}_{jt}}, & \text{if } j \in U_{i,t} \\ \text{NaN}, & \text{otherwise} \end{cases} \quad (1)$$

where tilt_{ijt} is tilt of investor i on industry j at quarter t . Holding_{ijt} represents the dollar value of investor i allocated to industry j at time t , and AUM_{it} is the total assets under management for investor i at time t . The second term captures the market capitalization (ME) weight of industry j relative to the total market capitalization across all industries. That is, I divide the market equity of each industry by the total market equity across all industries.⁴

Next, I present summary statistics for tilts. First, I construct the universal distribution of tilts by treating each investor-industry-quarter as a single observation and computing the overall distribution. Figure 3 depicts the resulting distribution. The tilt distribution exhibits three key features. First, it is leptokurtic, with a sharp peak around zero and heavy tails, indicating that while most investors hold neutral tilts, a fraction takes extreme positions. Second, its central tendency is near zero (mean: 0.002, median: -0.002), confirming neutrality on average. Third, its skewness appears minimal, though slight deviations may reflect biases in industry allocations.

The previous distribution treats each investor-industry-quarter as an independent observation, ignoring heterogeneity across these three dimensions. The following paragraphs explore this heterogeneity.

First, I examine the variation in tilt across investors. Figure 4 depicts the distribution of the sum of absolute tilts per investor, which exhibits a right-skewed shape, indicating

⁴Here, I have not restricted the denominator to the sum of industries within the investment universe of investor i . I am uncertain whether this is the correct approach.

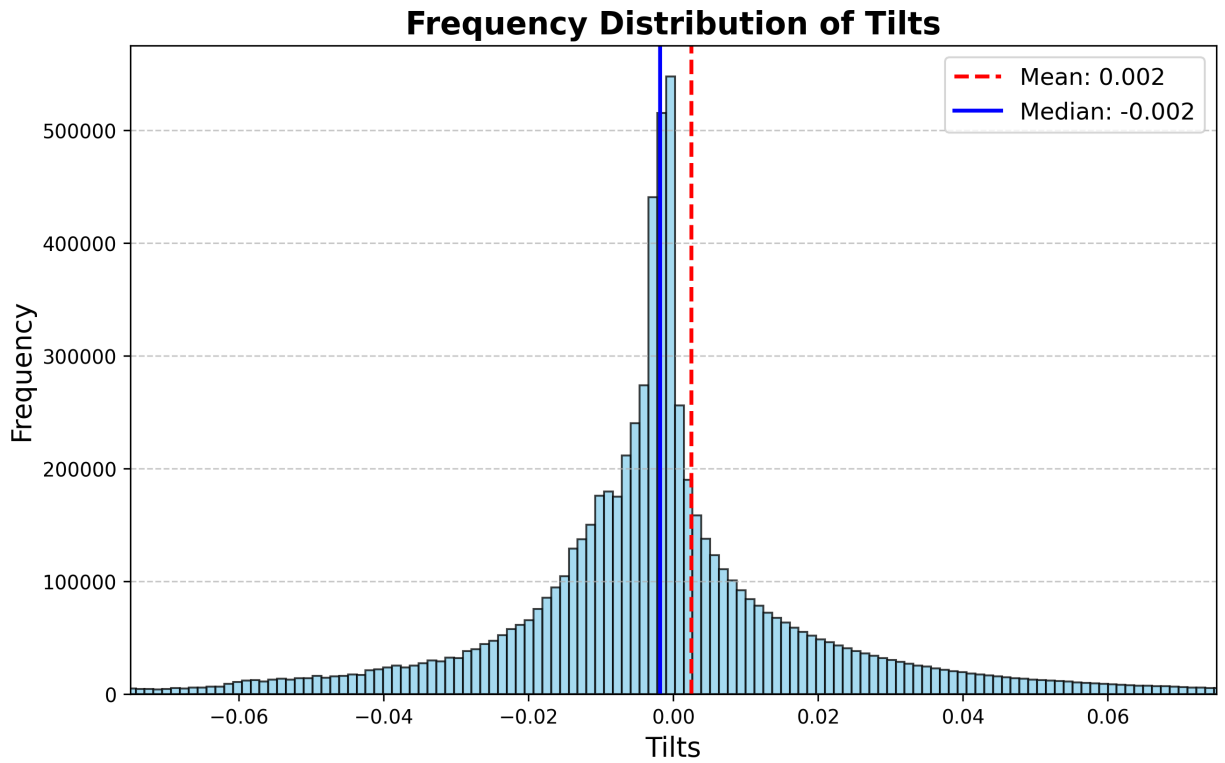


Figure 3: This figure shows the distribution of tilts across all investor-industry-quarter observations. I have cut the extreme tails for the sake of convenience in interpretations.

Distribution of Sum of Absolute Tilts per Manager across industries-quarter

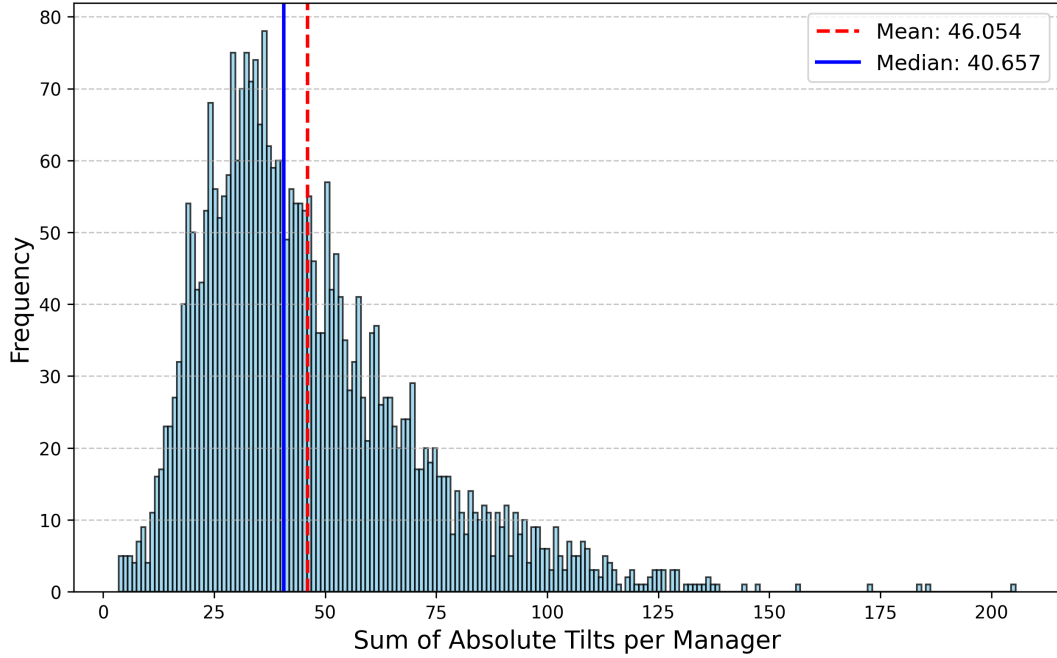


Figure 4: This graph depicts distribution of sum of absolute tilts across investors. For each investor, I drive sum of absolute value of tilts across industries and quarters and then plot it for all investors.

that while most managers maintain moderate levels of industry deviations, a subset exhibits significantly higher levels of tilts. The mean (46.054) is higher than the median (40.657), reflecting the influence of investors with exceptionally large tilts on the overall distribution. The peak near 40-50 suggests that most managers exhibit a relatively constrained deviation from the market portfolio, while the long tail extending beyond 100 highlights a few investors who take aggressive bets across industries. This distribution suggests heterogeneity in portfolio strategies, where some investors remain close to market weights while others actively overweight or underweight industries to a greater extent.

Next, I examine the behavior of aggregate tilt over time. To do so, I ignore heterogeneity across investors and industries and, for each quarter, compute the sum of the absolute values of tilts across all investor-industry pairs. To account for changes in the number of institutions, I divide this sum by the number of investors. Figure 4 presents this time series, where each point represents the sum of absolute tilts across investor-industry pairs, divided

by the number of industries in that quarter.

Conceptually, this trend can be decomposed into two components. First, it may signal market undervaluation, where informed investors tilt positively while uninformed investors tilt negatively. Here, the sum of the former and the absolute value of the latter drive the overall measure. Second, it may indicate a bubble, where informed investors underweight an industry—shifting toward outside assets—while uninformed investors overweight it.

The observed time series show distinct periods of rising and falling industry tilts, reflecting the dynamic nature of investor belief formation. The 1980s and early 2000s witnessed an increase in aggregate tilts, suggesting that institutions were either acting on relevant private information or responding to fads, narratives, or speculative trends. The sharp declines in tilts, particularly around 2000 (dot-com crash) and 2008 (financial crisis), suggest that market downturns coincide with investors reducing active bets, possibly due to increased uncertainty or a realization that prior deviations were excessive. If these tilts reflect rational information-based strategies, we would expect higher tilts to be associated with greater subsequent return predictability at the industry level.

If industry tilts capture investors belief shifts, they may have predictive power for aggregate market movements—contingent on identifying informed investors from uninformed ones. Periods of higher informed investors' tilts might indicate an environment where they expect higher market returns.

The extent to which this measure predicts market returns depends on whether tilts stem from relevant information or behavioral biases. If tilts reflect superior investor insights, higher tilts should precede positive excess returns. However, if they capture misguided sentiment, periods of high tilts may be followed by mean-reverting corrections as mispricing is resolved. A formal predictability test, such as regressing market returns on past tilt values, or a variance decomposition analysis, would help determine whether institutional tilts contribute to market efficiency or merely reflect noise and sentiment-driven misallocation ⁵.

⁵An exercise I am thinking to do

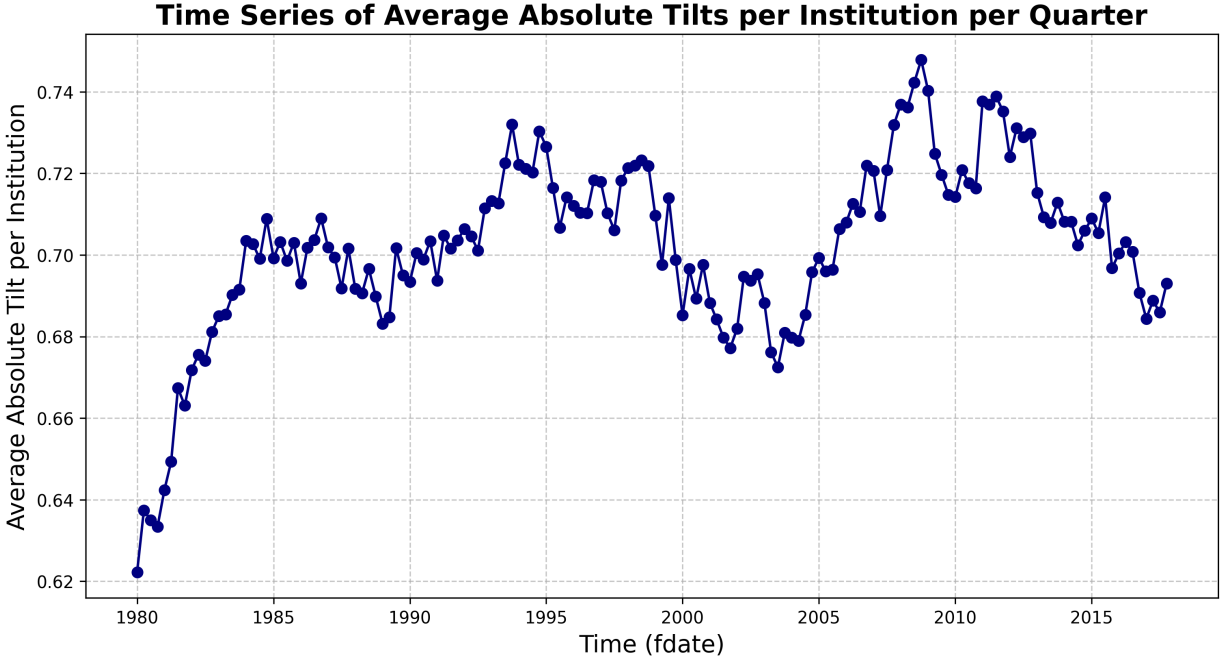


Figure 5: This graph shows the time series of aggregate tilt. At each quarter, I drive sum of absolute value of tilts across all investors and industries and then divide it by number of industries.

Next, I move toward heterogeneity of tilts across industries. Figure 6 ranks industries by the sum of absolute tilts, revealing which industries investors actively overweight or underweight relative to the market. The industries with the highest tilt values—Business Services, Financials, Oil, Pharmaceuticals, and Banks—are those where investors engage in the most significant deviations from market weights. These sectors likely attract more attention due to higher information asymmetry, evolving market trends, or macroeconomic sensitivity. Financial institutions and energy markets, for example, are highly responsive to interest rate changes and geopolitical risks, respectively, which may drive investors shifts. Similarly, pharmaceuticals are subject to regulatory approvals and innovation cycles, requiring sophisticated analysis that leads to stronger investors positioning.

On the other end, low-tilt industries such as Coal, Fabricated Products, Soda, and Guns see the least deviations, suggesting that investors either treat these industries as passive market allocations or lack a strong informational edge to make active tilts. Many of these sectors

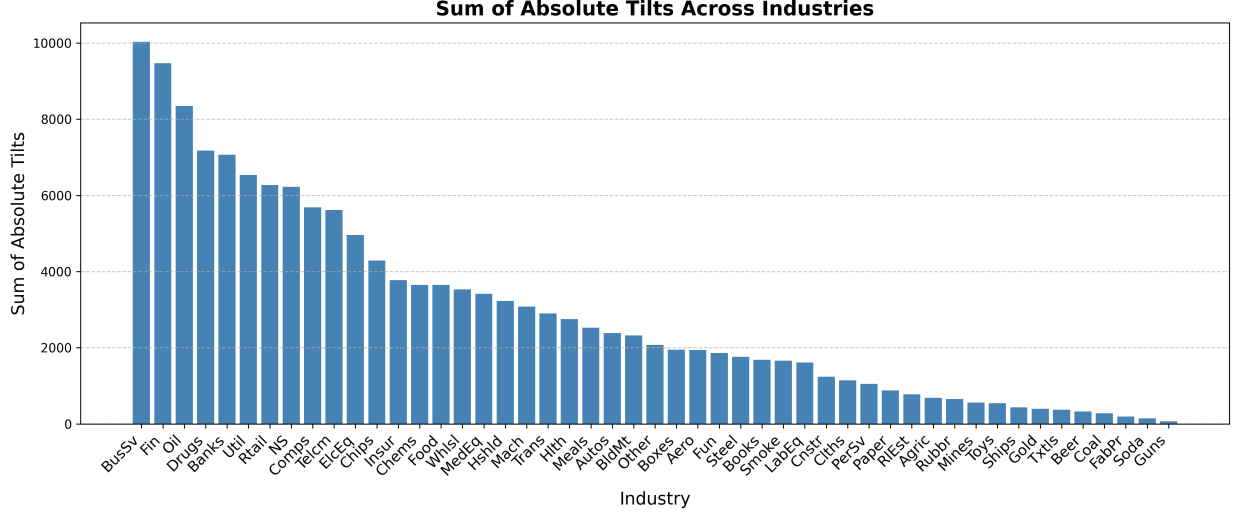


Figure 6: This graph shows the heterogeneity of tilts on industries. For each industry, sum of absolute values of tilts across investors and quarters are computed and the depicted.

are either highly stable with low volatility (e.g., Consumer Staples like Soda) or structurally declining (e.g., Coal and Guns), reducing incentives for investors to actively deviate from the market. Their limited tilt levels suggest less conviction, fewer arbitrage opportunities, or constraints (e.g., ESG investing restrictions on certain industries like coal and firearms). This asymmetric distribution of tilts highlights how investors attention clusters around dynamic, high-information sectors while remaining passive in traditional or declining industries.

So far, I have demonstrated significant heterogeneity in tilt across investors, industries, and time. Now, I examine the persistence of tilts in greater detail—a crucial aspect for my time series regressions. To do this, I compute the distribution of autocorrelations across investor-industry pairs, once again ignoring heterogeneity across investors and industries. Figures 7 and 8 depict the persistence of tilts over one-quarter and four-quarter horizons. The lag-1 autocorrelation distribution shows that most investor-industry pairs exhibit highly persistent tilts, clustering between 0.75 and 1, indicating that institutional allocations to industries remain relatively stable from quarter to quarter. However, there is notable variation, with some pairs displaying lower or even negative autocorrelations. This heterogeneity suggests a dual investor behavior: long-horizon investors, who maintain steady industry alloca-

tions, and active traders, who adjust tilts more frequently—potentially due to informational advantages—sometimes even reversing their positions in the short run.

Moving from lag-1 to lag-4, in figure 8, the persistence of tilts declines significantly across most investor-industry pairs, with the distribution shifting leftward and becoming more dispersed. This suggests that, while many institutions maintain their industry positions for a quarter or two, rebalancing becomes more pronounced over a year. Importantly, the degree of persistence varies across different investor-industry combinations—some remain highly stable, while others show negative autocorrelations at lag-4, indicating a full reversal of tilts. This pattern suggests that certain investors might be reacting to cyclical factors, macroeconomic changes, or correcting previous misallocations over longer horizons. The presence of both strongly persistent and mean-reverting tilt behaviors across different investor-industry pairs points to heterogeneous investment horizons, belief updating speeds, and institutional constraints, which contribute to the dynamic nature of industry capital allocation over time.

Figure 9 extends the previous analysis by visualizing heterogeneity across investors, industries, and time simultaneously. It presents a box plot of tilt autocorrelations across investors for different industries. Industries are sorted by their median persistence from left to right. This approach highlights substantial cross-industry variation in how persistently investors maintain their tilts.

Industries on the right side of the figure exhibit higher median autocorrelations, suggesting that investors in these sectors maintain stable tilts over time. This pattern indicates the presence of industry specialists or long-term strategic allocations, where investors hold strong sector-specific convictions. Notable examples include Coal, Firearms, and Gold, industries likely dominated by investors with deep expertise, leading to persistent and enduring capital allocations. The narrow interquartile ranges (IQRs) in these industries further reinforce that most investors exhibit similar persistence patterns, supporting the hypothesis that some industries attract dedicated, informed investors. Conversely, industries on the left side of the figure exhibit lower median autocorrelations and greater dispersion, implying

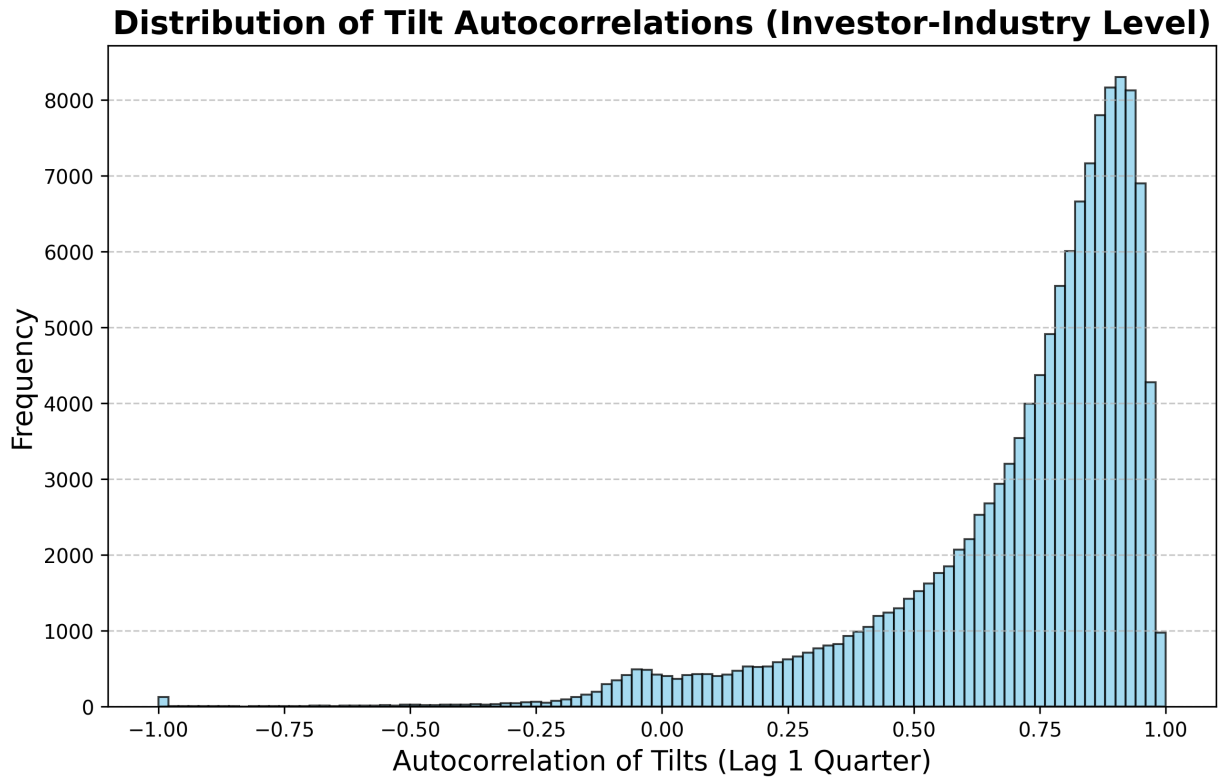


Figure 7: This graph shows the heterogeneity in persistence of tilts. For each pair of investor-industry first lag autocorrelation in tilts is computed and plotted.

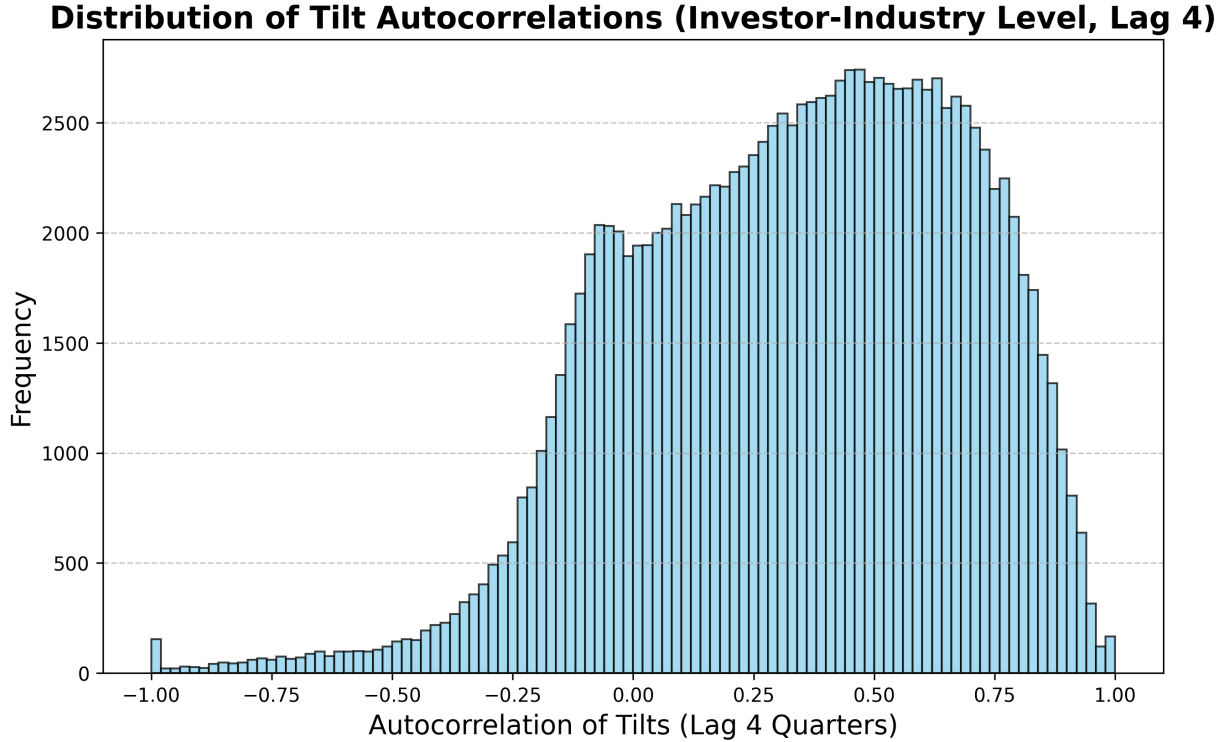


Figure 8: This graph shows the heterogeneity in persistence of tilts. For each pair of investor-industry 4th lag autocorrelation in tilts is computed and plotted.

higher turnover and frequent rebalancing of tilts. Sectors such as Business Services, Banking, and Other Financials show greater tilt variability, suggesting that investors in these industries frequently adjust their positions—likely in response to short-term information, earnings announcements, or macroeconomic shocks. The presence of negative autocorrelations and wider whiskers in these sectors indicates that some investors reverse their tilts over time, possibly following mean-reverting trading strategies or reflecting uncertainty in sector exposure decisions.

Additionally, the dispersion of points within each industry highlights investor-level heterogeneity, further emphasizing that different investors follow distinct strategies even within the same industry. This finding underscores the contrast between long-term industry specialists and short-term, opportunistic traders, illustrating how different sectors attract varying investment behaviors.

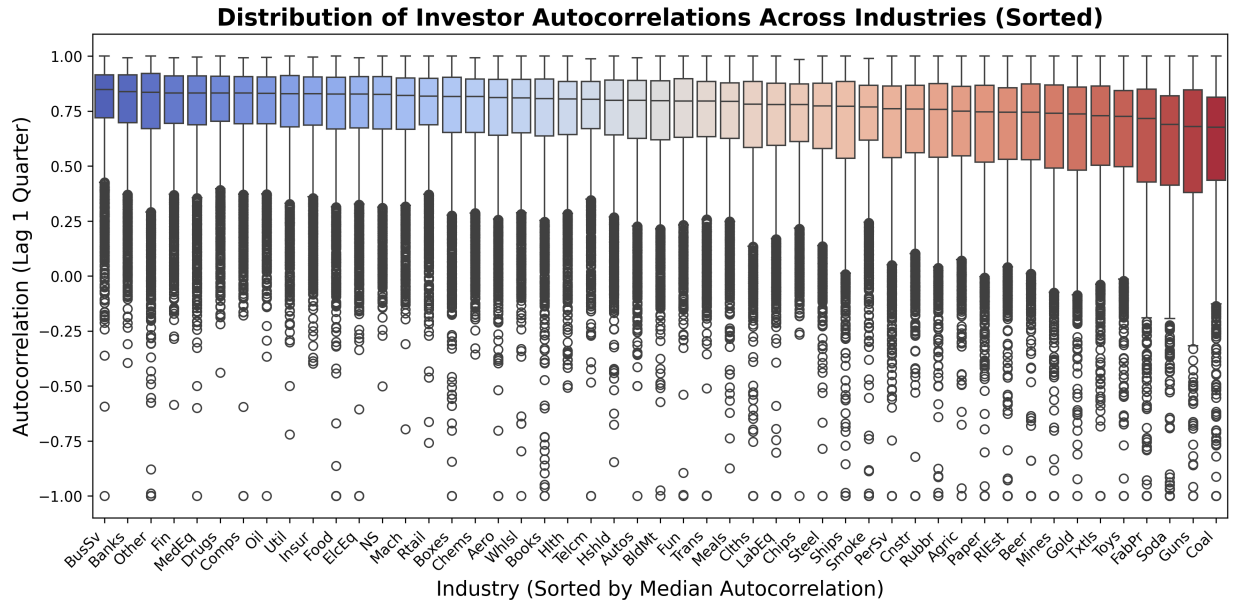


Figure 9: This figure shows the box-plot of autocorrelation for different industries. For each industry distribution of tilt across investors-quarters is plotted in the demonstrated box.

3 Empirical Framework and Results

In order to estimate distribution of industry returns from observed historical demand shifts, there are several possible methodologies. Using simple regressions is the most straightforward one which is the focus of this draft. Bayesian Updating, Machine Learning-Based Approaches, and Maximum Entropy Approach are another alternative approaches I am developing gradually. Below, I will elaborate on the first method and open the discussion for the others.

3.1 Regression Method

So far, I have highlighted key empirical patterns. First, a significant portion of investors concentrate their portfolios in a limited number of industries. Second, tilts exhibit heterogeneity across investors, over time, and across industries.

This section outlines the empirical methodology used to approximate information embedded in investors demand. Every quarter each industry receives a set of signal from different investors. The empirical results follow three steps. First, to define a mechanism which measure accuracy of each signal. Section 3.1.1 elaborates on this. Then, in section 3.1.3, I introduce some aggregation methods to aggregate signals across investors and come up with industry level signals. Finally, in section 3.1.3 I test to see if this aggregate signal predicts cross section of industries.

3.1.1 Signal Accuracy

At the end of each quarter, investors reveal their signal. Therefore, industry j at time t receives $S_{j,t} < I$ signals. $S_{j,t}$ is the number of investors which industry j is in their investment universe. The end goal is to aggregate this S signals across investors and come up with an industry level signal. To do this, one straightforward methodology is to drive weighted average of these signal, with the weight being a measure of each signal accuracy. So

that, the signals with higher accuracy rate have a more important effect on the prediction. Therefore, one first step is to define a mechanism to quantify accuracy of each signal.

This section ties accuracy of each signal to its sender. The logic is that accuracy of an observed signal defined as an average of how accurate the signal has been historically. Therefore, a simple way is to evaluate co-movement between historically available signal and future realized return. In other words, to define historical errors of the signal in predicting future returns.

To do this- drive accuracy of signal s observed at time t from investor i on industry j , or $s_{i,j,t}$ - I perform time series regressions as follow:

$$r_t = \alpha_{i,j,t} + \theta_{i,j,t}S_{t-1} + \gamma X_{t-1} + \varepsilon_t, \quad (2)$$

where:

- r_t is a vector of realized return from $t-k$ to t . With k being the length of a window i use. I use both fixed 30 quarters window or an expanded 30 quarters window.
- S_{t-1} is a window of historical signals up to time $t-1$. In other words, I argue that demand reveals expectations and therefore the current signal should predicts next period returns.
- X_{t-1} includes industry-level control characteristics including size, profitability, investment and dividend to book value of industries.
- $\theta_{i,j,t}$ is the estimated coefficient on signal from investor i in industry j at time t .

The interpretation of this regression is twofold. Assuming X represents the current information set, a one-unit change in investor i 's demand for industry j is expected to correspond to a θ change in its future return. Thus, θ captures the effect of investor i 's orthogonal information on industry j . The term "orthogonal" is preferable to "private" since the information may not be strictly private but rather undiscovered by econometricians. Additionally,

θ reflects demand shifts driven by expectations of future prices or other correlated factors. Regardless of the underlying motive, its predictive power for returns supports the paper’s objective—forecasting price movements.

With the first 30 quarters used as training data, the dataset spans from 1987 to 2017, covering a total of 120 quarters. Each period includes 49 industries and a varying number of investors, approximately 3,000, resulting in about 3.6 million signals in total.

Figure 10 presents the distribution of t-statistics for signals in the panel. Overall, slightly more than 10 percent of all received signals—approximately 370,000—are identified as containing useful information, while around 90 percent are classified as uninformative.

The bottom panel illustrates the distribution of significant coefficients, with extreme tails removed for clarity. This adjustment enhances readability and highlights the central tendency of the distribution.

After establishing an overview of accurate signals, I now interpret their conceptual meaning. A positive θ indicates that, after controlling for a set of characteristics, investor i ’s demand movements are systematically associated with future realized returns. This suggests two possible explanations. First, demand may be driven by true expectations, implying that investor i possesses orthogonal information about the next quarter’s industry price. This could stem from either access to latent information or a unique way of processing the current information set, which itself is perceived as an informational advantage.

An alternative explanation—that investor demand is motivated by hedging and systematically correlates with returns—seems unlikely. If demand were driven by hedging motives, the increased demand for risk protection should elevate asset prices, thereby reducing expected returns. Therefore, the most plausible interpretation of a positive θ is that the investor has a more accurate belief about the distribution of industry prices, translating into possession of orthogonal information.

The interpretation of a negative θ differs. A negative coefficient implies that the investor persistently predicts industry returns in the wrong direction. In other words, their demand

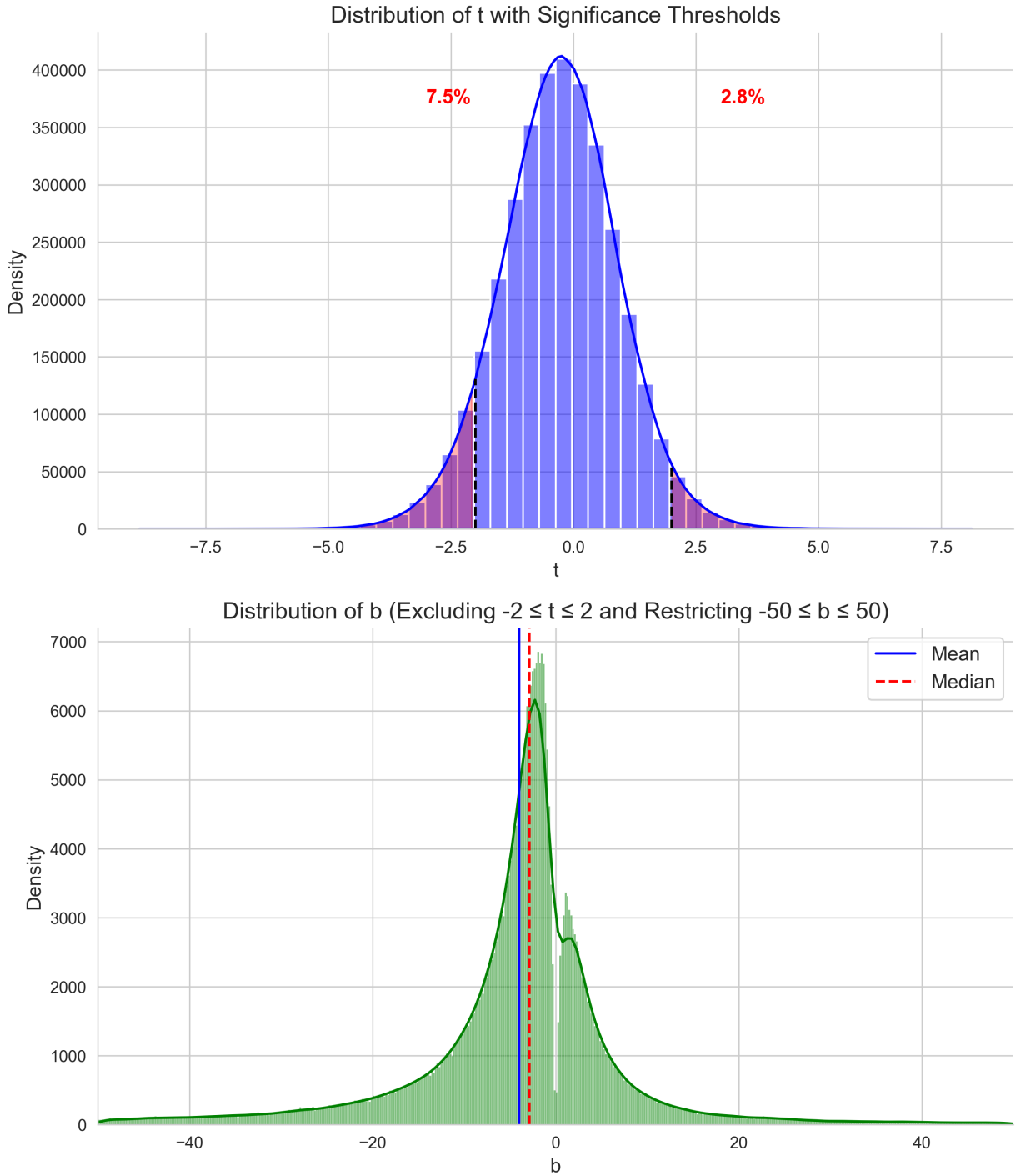


Figure 10: This figure plots distribution of all signals t statistics in the top panel and significant coefficients in the bottom panel.

changes are systematically followed by opposite price movements—suggesting that they buy assets just before a price decline.

Another possible explanation for a negative θ is that lower returns are associated with lower risks, leading to a rational demand for safer assets. This could reflect either risk aversion or non-pecuniary preferences. However, the latter is less likely since fundamental preferences, such as those driven by sustainability or ethical considerations, do not typically fluctuate on a quarterly basis.

A zero coefficient indicates that an investor’s demand dynamics are not informative about future prices. This could be because their demand remains stable over time—such as long-horizon investors following a buy-and-hold strategy—or because their information set lacks predictive value, effectively making their signals indistinguishable from noise.

For clarity and alignment with economic and statistical interpretations, I label these signals into three distinct groups. Investors whose demand shifts provide no predictive power are labeled as ****uninformative****, or “noise” signals, consistent with prior literature. Investors who systematically predict returns in the wrong direction are labeled ****misinformative****, or “inverse” signals. Lastly, investors whose demand shifts accurately predict future returns are termed ****informative****, or “industry experts.”

Mathematically, each signal is classified into one of the three following signals:

$$s_{i,j,t}^{informative} = \mathbb{1}(t\text{-stat}(\theta_{i,j,t}) > 2).$$

$$s_{i,j,t}^{misinformative} = \mathbb{1}(t\text{-stat}(\theta_{i,j,t}) < -2).$$

$$s_{i,j,t}^{uninformative} = -2 \leq \mathbb{1}(t\text{-stat}(\theta_{i,j,t}) \leq 2).$$

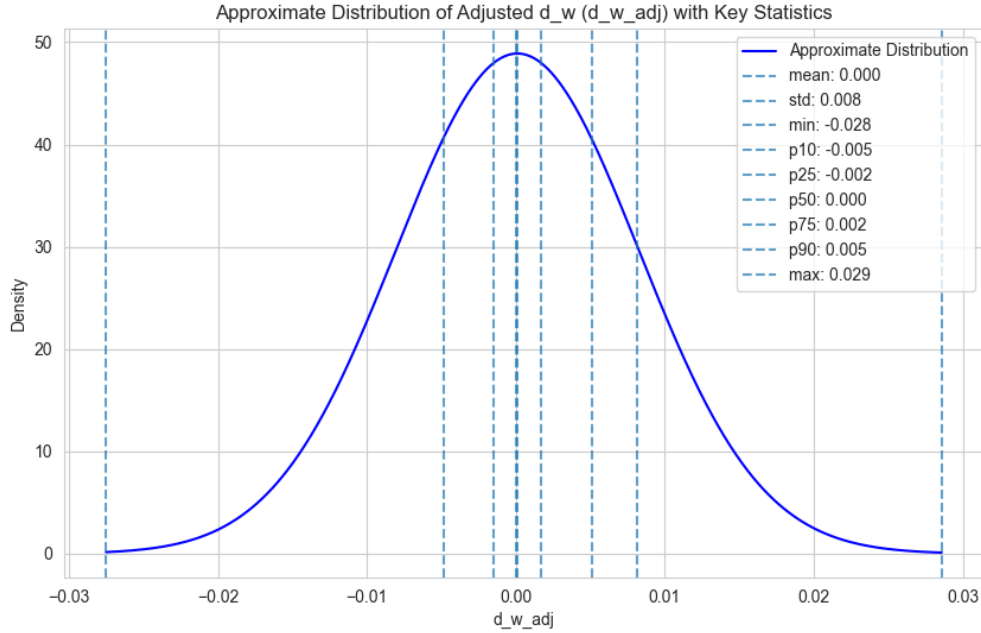


Figure 11: This figure shows an approximation of distribution of signals across industry-quarter distributions.

Figure 11 shows an approximation of distribution of signals across industry-quarter distributions.

To enhance the informativeness of signals, I define an investor as informative or misinformative if they have predicted an industry in at least five quarters. This criterion does not impose a restriction on the dataset but serves as a principle in the training algorithm.

Having established the interpretation of the coefficients, I now examine key characteristics of these three types of signals. One particularly relevant dimension is portfolio diversification among investors. Specifically, I analyze how many industries they hold and in how many industries they are classified as informative, misinformative, or uninformative.

Table 2 reports descriptive statistics for these investor groups, with and without the informativeness restriction. The remainder of the analysis is conducted under the restricted classification framework.

After applying the five-quarter requirement, approximately 9 percent of the 3.6 million

signals are classified as informative. Among them, 2.1 percent exhibit a direct relationship with price, while 6.5 percent are associated with reverse predictions.

The third row in Table 2 presents the number of managers. These signals originate from 3,023 distinct investors. About 66 percent of investors have sent at least one informative signal, while approximately 77 percent have been identified as misinformed at least once. Notably, all investors have sent at least one uninformative signal.

The second part of Table 2 provides further insight into the nature of these investors and the sources of the signals. A few notable observations emerge. First, 90 percent of the accurate signals originate from investment advisors. In contrast, household investors predominantly generate reverse signals, indicating that their demand shifts are negatively correlated with stock returns. Additionally, insurance companies, banks, and pension funds are slightly more likely to be classified as reverse predictors.

Table 2: Summary statistics of investor signals, categorized as informative, misinformative, and uninformative. The table reports the number of signals, distribution across investor types, and the prevalence of each signal classification.

	All	Uninformative	Informative	Misinformative
Before				
#	3,559,690	3,193,687	99,000	269,255
%	100%	89.7%	2.8%	7.6%
After				
#	3,559,690	3,252,812	76,324	230,825
%	100%	91.4%	2.1%	6.5%
investor				
#	3,023	3,023	1,991	2,317
%	100	100	66	77
Type Distribution (%)				
Investment Advisors		47.92	41.73	48.68
Mutual Funds		30.00	23.17	25.36
Banks		11.99	19.65	14.82
Insurance Companies		4.42	6.57	5.34
Pension Funds		3.13	6.40	3.26
Other		2.48	1.92	2.41
Households		0.05	0.55	0.13

Table 3 presents summary statistics of signals across managers. The dataset includes 3,023 managers, with each sending an average of 1,261 signals. However, the distribution varies widely, with some managers submitting as few as one signal, while others have sent nearly 10,000.

On average, each investor has contributed 1,261 signals, of which 1,153 are classified as uninformative, 81 as misinformative, and 27 as informative.

Table 3: Summary statistics of signals across managers. The table reports the total number of investors, the distribution of signals per investor, and the average classification of signals as informative, misinformative, or uninformative.

	Total Signals	Uninformative	Informative	Misinformative
Count	3023	3023	3023	3023
Mean	1261	1153	27	81
Std Dev	1386	1250	40	130
Min	1	1	0	0
25th Pct	256	243	0	5
50th Pct	763	711	11	34
75th Pct	1772	1617	36	102
Max	9654	8882	312	1342

Table 4 provides a deeper analysis of investor classifications. Thus far, no distinction has been made between investors who exclusively send informative signals and those who send both informative and misinformative signals. Each investor is now categorized as always informative, always misinformative, or a combination—meaning they are informative in some industries, uninformative in others, and in some cases, both within the same industry at different times.

Overall, 18 percent of investors have only sent uninformative signals, 16 percent have exclusively sent misinformative signals, and 5 percent have only sent informative signals. The remaining 61 percent of investors have sent both types of signals.

The second row in Table 4 reports statistics on portfolio diversification, measured by the number of industries in which investors hold positions. Investors who never send informative signals have, on average, 20 industries in their portfolios. Informative investors are slightly more diversified, while misinformative investors hold an average of 31 industries. Investors who send both signals are the most diversified, with an average of 40 industries in their portfolios.

The third section of the table describes the number of industries in which investors are classified as experts. On average, misinformative investors send misinformation in 3.5 industries, while informative investors are experts in just over one industry. Investors classified as both informative and misinformative are experts in an average of 10 industries.

Among the 3,023 investors, 1,980 fall into the "both" category, meaning that most investors are experts in some industries while systematically making errors in others. There are 150 investors who are experts in at least one industry. Their diversification patterns are also notable: experts invest in an average of 23 industries but are only classified as experts in 1.4 of them. In contrast, novices have more diversified portfolios, holding positions in 31 industries while making systematic mistakes in 3.5 industries on average. Interestingly, neutral investors, who do not send informative signals, are the least diversified. Investors classified as both informative and misinformative tend to be the most diversified, with expertise in an

average of 12 industries.

Table 4: Classification of investors based on their signal informativeness. The table reports the distribution of investors by signal type, their portfolio diversification, and the number of industries.

	Uninformative	misinformative	informative	Both
#	556	476	150	1841
%	18	16	5	61
Industries in portfolios				
Mean	20	31	24	40
Median	19	31	23	41
Std	12.13	9.99	10.23	8.00
Industries they are informative,...				
Mean	0.00	3.47	1.40	12.17
Median	0	2	1	10
Std	0.00	3.25	0.80	8.47

Table 5 provides further statistics on the 1,841 investors who send mixed signals. These investors are classified as experts in some industries, novices in others, and, in some cases, exhibit both correct and inverse predictions for the same industry at different times.

The first column of the table presents statistics on the number of industries in which these investors are identified as experts. On average, they are experts in more than three industries, with the total number of expert-industry pairs reaching approximately six per investor. Additionally, they are classified as misinformative in over eight industries on average, amounting to approximately 15,000 industry-investor observations.

Among these mixed-signal investors, 556 have been identified as both informative and misinformative for at least one industry. In total, 1,137 industry-investor pairs fall into this category, with an average of 0.62 per investor. However, no investors have been identified as informative in both categories without also being classified elsewhere.

Lastly, the number of industries that have received both informative and misinformative signals from the same investor is statistically insignificant, indicating that mixed-signal investors tend to exhibit systematic biases rather than random fluctuations in signal type.

Table 5: Summary statistics of investors who send mixed signals. The table reports the number of industries in which these investors are classified as experts or misinformative, as well as instances where both signal types appear for the same industry.

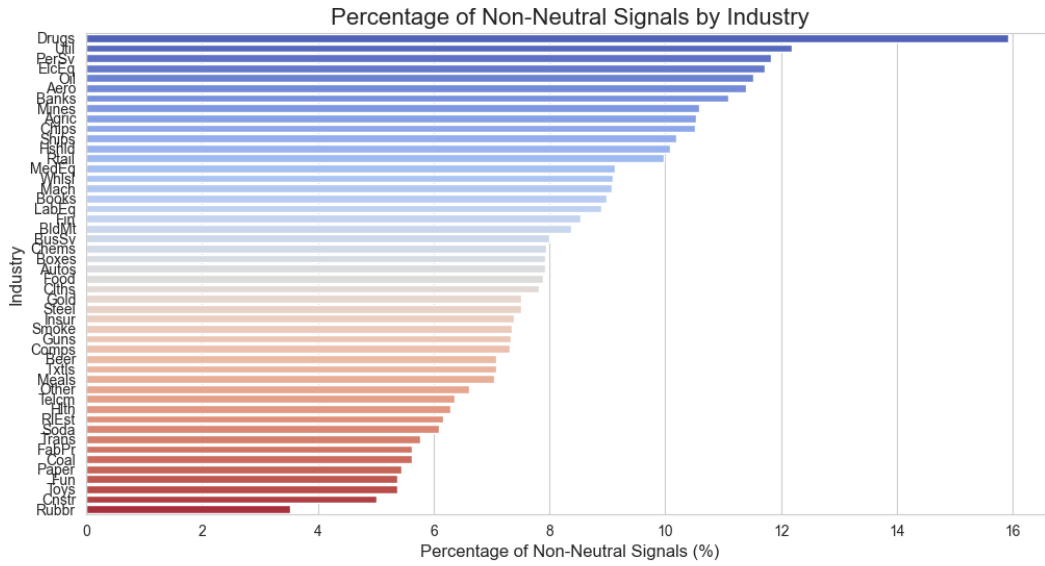
	Informative	misinformative	both
Count	1823	1841	556
Mean	3.25	8.30	0.62
Std Dev	2.41	6.75	1.27
Min	0.00	1.00	0.00
25th Pct	1.00	3.00	0.00
50th Pct	3.00	7.00	0.00
75th Pct	4.00	11.00	1.00
Max	18.00	43.00	9.00
Number of Industries	5984	15289	1137
Percent	26.70	68.22	5.07

Next, I analyze the distribution of signals across industries. At any given point in time, each industry receives a large number of signals. Two key questions arise: Do different industries receive similar proportions of uninformative signals? And do industries differ in the relative share of informative and misinformative signals they receive? The answers to these questions may serve as a proxy for the level of uncertainty in an industry’s future price movements.

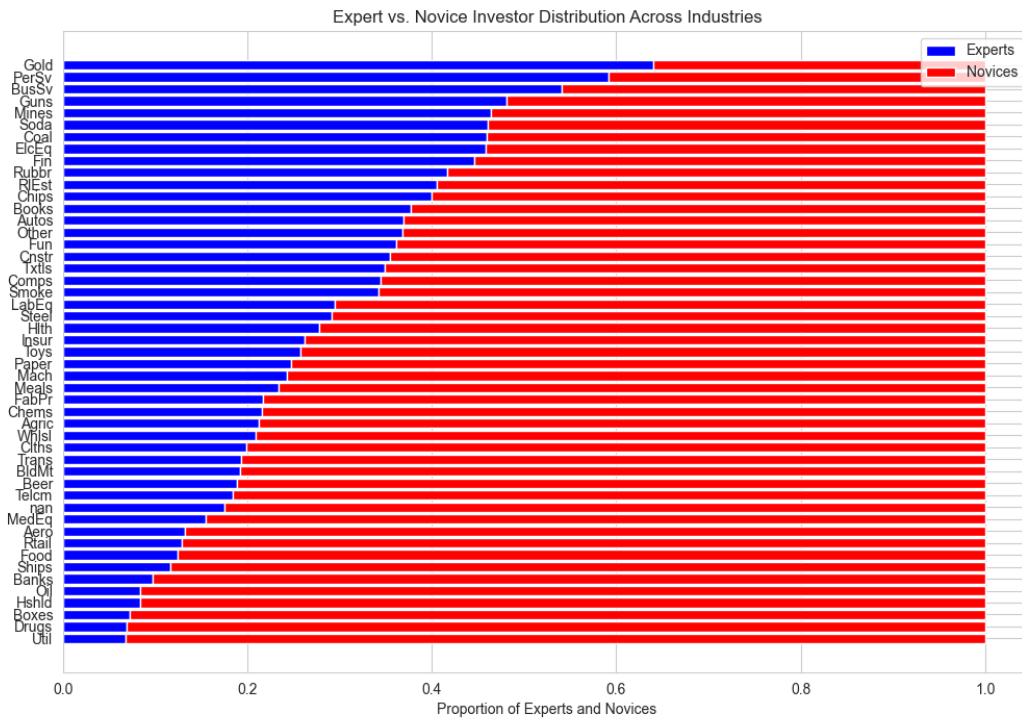
The top panel of Figure 12 presents the fraction of non-neutral signals—both informative and misinformative—relative to all signals received by each industry. The variation across industries suggests that certain industries attract a larger share of signals containing potential information, whereas others predominantly receive uninformative signals. This discrepancy may reflect differences in industry-specific uncertainty, information asymmetry, or investor attention.

The bottom panel further dissects the composition of these non-neutral signals by distinguishing between informative signals (experts) and misinformative signals (novices). Industries differ significantly in the proportion of expert and novice signals, indicating that some industries are more prone to systematic mispredictions, while others are characterized by a greater presence of informed investors. This pattern may serve as a proxy for the level of uncertainty in an industry’s future price movements, with industries dominated by novice signals potentially experiencing greater mispricing or volatility.

Changes in demand can occur through two primary channels. First, the intensive margin, where one asset is sold and another is purchased, results in a positive demand on the first asset and a negative demand on the second one. Second, the extensive margin, where new capital inflows occur, does not affect tilts if the inflow is proportional across assets. However, if the new capital is disproportionately allocated to a particular asset, a plausible explanation is a higher expected return on that asset.



(a) Percentage of non-neutral signals by industry



(b) Industry signal distribution

Figure 12: Comparison of industry signals. The top panel shows the percentage of signals which are informative or misinformative by industry, while the bottom panel presents the share of each for each industry

3.1.2 Signal Aggregation

Now that the accuracy of signals has been identified, this section focuses on aggregating signals to construct an industry-level measure of orthogonal information, or expected return. Before discussing aggregation methods, I first define the signal itself. Signal, is considered as change in portfolio weights.

$$s_{i,j,t} = \Delta w_{i,j,t} = w_{i,j,t} - w_{i,j,t-1} \quad (4)$$

Where

$$\frac{\text{holdings}_{i,j,t}}{\sum_{j \in F_t} \text{holdings}_{i,j,t}} \quad (5)$$

where $w_{i,j,t}$ represents the fraction of investor i 's portfolio allocated to industry j at time t , and $\text{holdings}_{i,j,t}$ is the total dollar value of investor i 's holdings in industry j . The denominator ensures that the weight is computed relative to the investor's total portfolio across all industries F_t .

This captures how much investor i adjusts their exposure to industry j from the previous quarter. A positive Δw implies an increase in expected return- or other demand motives- by the investor, whereas a negative Δw implies a decrease. However, I assume the sign of θ in times series regression separates expected returns motives from other demand motives. That is positive coefficient investors' change in demand is due to a higher expected return.

Before delving into aggregation, I provide some information on demand change or Δ Table 6 provides summary statistics for the three previously mentioned groups of investors. Several key insights emerge. First, the mean signal for informative investors is significantly lower than that of uninformative investors, indicating that informative investors adjust their portfolio positions more aggressively based on their beliefs. Second, the proportion of zero signals differs across groups—approximately 11.2 percent of informative and uninformative signals are zero, whereas only 6 percent of misinformative signals are zero. This suggests that misinformative investors tend to trade more actively, even when their signals are inaccurate.

Table 6: Summary Statistics of Investor Signals

Panel A: Distribution of d_w			
Statistic	Uninformative	Informative	Misinformative
Count	3,252,812	76,324	230,825
Mean	-0.00003	-0.00013	-0.00016
Std Dev	0.0187	0.0179	0.0121
Min	-1.0000	-1.0000	-0.9306
25%	-0.0014	-0.0016	-0.0016
50% (Median)	0.0000	0.0000	0.0000
75%	0.0015	0.0016	0.0015
Max	1.0000	0.7436	0.7936

Panel B: Number of Observations per Industry-Date		
Statistic	Informative	Misinformative
Count	5,591	5,858
Mean	13.65	39.40
Std Dev	19.64	45.28
Min	1.00	1.00
25%	4.00	11.00
50% (Median)	7.00	24.00
75%	15.00	51.00
Max	200.00	335.00

Notes: Panel A presents summary statistics for the distribution of d_w across three investor groups. Panel B provides the distribution of the number of observations per industry-date combination for the informative and misinformative groups.

Several methods can be used to aggregate signals into an industry-level measure. The simplest approach is to compute the unweighted mean across all signals within a given group. This method assumes that all investors in the selected group exhibit the same level of accuracy, thereby treating their signals as equally informative. Since each signal reflects an investor's change in expected returns for an industry, a higher signal corresponds to a higher expected return. The industry-level aggregation of informative signals, termed Revealed Industry Information ($\text{RII}_{\text{informative}}$), is defined as:

$$\text{RII}_{j,t}^{\text{informative}} = \frac{1}{N_{\text{informative}}} \sum_{i \in S_{j,t}^{\text{informative}}} \Delta w_{i,j,t}, \quad (6)$$

Similarly, the aggregated measure for misinformative investors is:

$$\text{RII}_{j,t}^{\text{misinformative}} = \frac{1}{N_{\text{misinformative}}} \sum_{i \in \mathcal{S}_{j,t}^{\text{misinformative}}} \Delta w_{i,j,t}, \quad (7)$$

where $\text{RII}_{j,t}^{\text{misinformative}}$ represents signals from investors who systematically mispredict future returns. Finally, investors whose signals do not exhibit any meaningful relationship with future returns are classified as uninformative, and their aggregated signal is:

$$\text{RII}_{j,t}^{\text{uninformative}} = \frac{1}{N_{\text{uninformative}}} \sum_{i \in \mathcal{S}_{j,t}^{\text{uninformative}}} \Delta w_{i,j,t}. \quad (8)$$

To improve aggregation accuracy, I also construct a weighted measure that assigns different weights to each signal based on the magnitude of the investor's coefficient in time-series regressions. This accounts for the varying degrees of informativeness among investors. The first weighted measure incorporates the estimated coefficient (θ) as a weight:

$$\text{RII}_{j,t}^{\text{informative, weighted}} = \frac{\sum_{i \in \mathcal{S}_{j,t}^{\text{informative}}} \theta_{i,j,t} \cdot \Delta w_{i,j,t}}{\sum_{i \in \mathcal{S}_{j,t}^{\text{informative}}} \theta_{i,j,t}}, \quad (9)$$

where $\theta_{i,j,t}$ represents the estimated coefficient of investor i 's signal.

A second approach applies the inverse of the coefficient as a weight:

$$\text{RII}_{j,t}^{\text{informative, inverse-weighted}} = \frac{\sum_{i \in \mathcal{S}_{j,t}^{\text{informative}}} \frac{1}{\theta_{i,j,t}} \cdot \Delta w_{i,j,t}}{\sum_{i \in \mathcal{S}_{j,t}^{\text{informative}}} \frac{1}{\theta_{i,j,t}}}. \quad (10)$$

To capture the combined effect of both informative and misinformative signals, I construct an adjusted Revealed Industry Information (RII) index. This measure incorporates the average demand shifts of informative investors, while also accounting for misinformed investors by taking the negative of their demand shifts. The intuition behind this adjustment is that misinformed investors systematically predict returns inversely, and their demand shifts should be reversed when constructing an aggregate expectation measure. The resulting index, RII_{net} , reflects the overall investor expectation while filtering out systematic

misinformation.

$$\text{RII_net}_{j,t} = \frac{1}{N_{\text{inf}} + N_{\text{misinf}}} \left[\sum_{i \in \mathcal{S}_{j,t}^{\text{inf}}} \Delta w_{i,j,t} - \sum_{i \in \mathcal{S}_{j,t}^{\text{misinf}}} \Delta w_{i,j,t} \right]. \quad (11)$$

These industry-level measures serve as a proxy for investor expectations about future returns. A high positive RII suggests that informed investors anticipate higher future returns, whereas low or negative RII implies lower expected returns. These measures provide an aggregated, market-based proxy for industry expectations.

3.1.3 Industry Return Predictability

So far, signals have been analyzed in terms of their accuracy and aggregated to construct a measure of industry-level latent information or expected return. I now examine whether $RII_{j,t}$ explains the cross-section of industry returns. The coefficient γ in the following Fama-MacBeth regression is expected to capture this relationship:

$$R_{j,t+1} - R_m = \alpha + \gamma RII_{j,t} + \beta X_{j,t} + \epsilon_{j,t} \quad (12)$$

where the left-hand side represents the realized return of industry j in the next period, and the right-hand side includes industry characteristics $X_{j,t}$ along with $RII_{j,t}$, which serves as the industry-level measure of expected return based on information available up to quarter t .

The coefficient γ captures the extent to which investors' revealed expectations, as measured by RII , explain the cross-section of future industry returns. A significant γ would suggest that investor demand shifts contain valuable information about expected returns.

Table 7 presents summary statistics for RII alongside control variables, where RII serves as the predictor and industry return as the predicted variable. The first five rows summarize the five constructed RII indices. The primary focus is on the first row, where the signal is the simple average of informative investors' signals. The reported statistics are calculated across 49 industries at each quarter, and the average across all quarters is displayed. The signal is scaled by 100 for ease of interpretation.

The $RII_{\text{informative}}$ index assigns a value between -2.14 and 2.76 to each industry, with a mean of 0.01 and a standard deviation of 0.72, suggesting meaningful variation in expectations. The same level of dispersion is observed in the other RII measures.

The $RII_{\text{misinformative}}$ index varies between -1.07 and 1.05, with a mean of 0.00 and standard deviation of 0.33. The weighted version of the informative index, $RII_{\text{inf.w}}$, has a mean of 0.04 and standard deviation of 0.48, while the inverse-weighted version, $RII_{\text{inf.w-}}$, has a mean

of -0.05 and a standard deviation of 1.54.

The next two columns report returns. The average quarterly return is 4.67%, with a standard deviation of 8.19%. The difference between the 10th and 90th percentiles is 18 percentage points, indicating considerable variation in realized industry returns.

The last row presents the RII_{net} index, which combines both informative and misinformative signals. This measure aggregates informative signals while subtracting the negative contribution of misinformed investors. As shown in the table, RII_{net} ranges from -8.8 to 8.7, with a mean of -0.02 and a standard deviation of 2.82.

Table 7: Summary Statistics of Key Variables

Variable	Mean	Std Dev	Min	10th	25th	50th	75th	90th	Max
$RII_{\text{informative}}$ (x100)	0.10	7.20	-21.40	-4.50	-1.50	-0.10	1.30	4.30	27.60
$RII_{\text{misinformative}}$ (x100)	-0.00	3.30	-10.70	-2.80	-0.90	0.00	1.00	2.70	10.50
$RII_{\text{uninformative}}$ (x100)	-0.00	2.10	-6.80	-1.90	-0.70	-0.00	0.70	1.90	6.20
$RII_{\text{inf.w}}$ (x100)	0.40	4.80	-10.90	-2.40	-0.80	0.00	0.80	2.90	21.00
$RII_{\text{inf.w.inv}}$ (x100)	-0.50	15.40	-58.30	-7.90	-2.40	-0.10	1.80	7.10	49.50
RII_{net} (x100)	-0.02	2.82	-8.80	-2.38	-0.90	-0.01	0.87	2.32	8.73
Market Equity (me /1000)	215.14	257.86	2.61	14.04	32.94	117.07	266.50	604.66	1081.40
divA.be	0.04	0.02	0.00	0.01	0.02	0.03	0.05	0.07	0.09
Profitability	0.29	0.10	0.02	0.17	0.23	0.29	0.35	0.40	0.54
Gat	0.11	0.09	-0.11	0.02	0.06	0.10	0.15	0.21	0.41
Return (ret x100)	4.67	8.19	-13.99	-4.09	-0.02	4.10	8.83	14.02	27.64
Six-Month Return (ret_6 x100)	9.70	12.46	-17.52	-3.41	2.52	8.73	15.80	23.56	46.97

Notes: This table presents summary statistics for key variables. The RII measures are scaled by 100 for readability. Market equity (me) is reported in thousands. The six-month return (ret_6) is adjusted by first subtracting 1 and then multiplying by 100 to reflect percentage returns. The variables divA.be, profitability, and Gat are in their original units.

The regression results for different variations of the Revealed Industry Information (RII) signal are reported in Table 8. Three variations are considered: unweighted signals, weighted signals, and combined signals. The first variation includes the simple average of informative investors' signals and misinformative investors' signals. The second variation weights signals based on the estimated accuracy of each investor's demand shift. The third variation combines both informative and the negative of misinformative signals.

Each regression examines whether RII explains the cross-section of future industry returns. The first column reports the results when using `IEE_informative`, which is the average demand shift of investors whose demand positively correlates with future industry returns. The coefficient is 1.05, meaning a one-unit increase in `IEE_informative` is associated with a 1.05-unit increase in future industry returns. A one-standard-deviation increase in `IEE_informative` translates to a 75-basis-point increase in quarterly returns.

The second column tests whether `IEE_misinformative`, the average demand shift of investors whose demand is negatively correlated with future returns, predicts returns. The coefficient is -1.04, meaning a one-unit increase in demand from misinformative investors is associated with a 1.04-unit decline in returns. A one-standard-deviation decrease in `IEE_misinformative` is associated with a 35-basis-point increase in quarterly returns.

The results for two weighted measures are shown in columns three and four. In column three, `IEE_informative_w` weights signals by their estimated accuracy. The coefficient is 1.88, meaning a one-standard-deviation increase in the weighted signal predicts a 90-basis-point increase in returns. Column four uses an alternative weighting based on the inverse accuracy, with a coefficient of 0.56, suggesting that weighting by inverse accuracy still predicts returns but with a smaller magnitude.

Column five reports the results when `IEE_informative_misinformative` is used. This regression aggregates both informative and the negative of misinformative signals. The coefficient is positive and significant, suggesting that both informative and misinformative investors provide independent predictive power for returns.

Column six tests IEE.uninformative. As expected, its coefficient is close to zero, implying that uninformative signals do not contain meaningful information for future returns.

Finally, column seven considers using the information of both informative and misinformative simultaneously. The coefficient on this combined measure is 2.16, implying that a one-standard-deviation increase in this index is associated with a 61-basis-point increase in quarterly returns. This suggests that aggregating both informative and misinformative signals enhances predictive power. In other words, misinformative signals provide independent predictive information.

Table 8: Regression Results when signal is Δw

$$R_{j,t+1} - R_m = \alpha + \gamma RII_{j,t} + \beta X_{j,t} + \epsilon_{j,t}$$

round-mode = places, round-precision = 2,

Factor	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RII_inf	1.05*** (0.30)	—	—	—	0.85*** (0.24)	—	—
RII_misinf	—	−1.04** (0.48)	—	—	−1.04** (0.45)	—	—
RII_inf_w	—	—	1.88** (0.79)	—	—	—	—
RII_inf_w_inv	—	—	—	0.56** (0.24)	—	—	—
RII_uninformative	—	—	—	—	—	−0.63 (0.72)	—
RII_inf_misinf	—	—	—	—	—	—	2.16*** (0.52)
divA_be	−0.23* (0.13)	−0.30** (0.13)	−0.05 (0.26)	−0.39** (0.19)	−0.26* (0.13)	−0.31** (0.14)	−0.24* (0.12)
me	-5.87×10^{-9} (9.57e-09)	-9.09×10^{-9} (1.25e-08)	1.90×10^{-8} (2.10e-08)	-3.06×10^{-8} (3.01e-08)	-7.95×10^{-9} (1.12e-08)	-8.11×10^{-9} (1.12e-08)	-8.04×10^{-9} (1.13e-08)
profit	0.02 (0.02)	0.03 (0.02)	−0.02 (0.05)	0.03 (0.02)	0.02 (0.02)	0.03 (0.02)	0.03 (0.02)
ret_3	0.02 (0.04)	−0.02 (0.04)	0.03 (0.05)	0.04 (0.05)	0.01 (0.04)	−0.01 (0.04)	−0.02 (0.04)
ret_6	0.02 (0.03)	0.05 (0.03)	−0.01 (0.05)	0.03 (0.03)	0.03 (0.03)	0.04 (0.04)	0.05 (0.03)
Gat	−0.02 (0.02)	−0.03* (0.02)	−0.00 (0.03)	−0.02 (0.02)	−0.03* (0.02)	−0.05** (0.02)	−0.03* (0.02)
Intercept	0.02 (0.04)	0.00 (0.03)	0.04 (0.04)	0.02 (0.03)	0.02 (0.03)	0.02 (0.04)	−0.01 (0.03)
Observations	[N ₁ , N ₂ , N ₃ , N ₄ , N ₅ , N ₆ , N ₇]						

Notes: Each column represents a separate regression. Standard errors are in parentheses. Dashes indicate

that a variable was not included in that regression. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

Table 9 shows the results when changes in investors' tilts, instead of change in portfolio weights, are interpreted as shifts in demand. As shown in the table, the coefficient on the mean change in tilts of industry experts significantly predicts industry returns. A one-unit increase in the mean change in tilt corresponds to a 1.6-unit increase in quarterly industry returns. When higher weights are assigned to signals perceived as more accurate, the coefficient becomes larger and remains significant. However, the mean tilt changes of those whose signals negatively predict industry returns do not significantly explain future returns.

A one-standard-deviation increase in IEEis associated with a 0.44 percentage point increase in returns, while the weighted measure implies a 0.58 percentage point increase.

Table 9: Regression Results when signal is $\Delta tilt$

$$R_{j,t+1} - R_m = \alpha + \gamma RII_{j,t} + \beta X_{j,t} + \epsilon_{j,t}$$

Factor	(1)	(2)	(3)	(4)	(5)	(6)
RII_inf	1.6693*** (0.4791)	—	1.4155*** (0.4935)	—	—	—
RII_misinf	—	−0.3786 (0.2981)	−0.3314 (0.3084)	—	—	—
RII_inf_w	—	—	—	2.8690*** (0.8741)	—	—
RII_inf_w_	—	—	—	—	0.4371*** (0.1529)	—
RII_inf_misinf	—	—	—	—	—	2.0186*** (0.5569)
Gat	−0.0427** (0.0197)	−0.0389** (0.0193)	−0.0382* (0.0202)	−0.0406** (0.0197)	−0.0369* (0.0195)	−0.0392** (0.0200)
divA_be	−0.2883** (0.1311)	−0.3097** (0.1331)	−0.2889** (0.1353)	−0.3195** (0.1326)	−0.3181** (0.1348)	−0.3192** (0.1342)
me	-3.51×10^{-9} (8.81e-09)	-2.90×10^{-9} (9.03e-09)	-3.74×10^{-9} (9.20e-09)	-2.21×10^{-9} (8.90e-09)	-4.18×10^{-9} (9.38e-09)	-3.41×10^{-9} (9.14e-09)
profit	0.0322 (0.0203)	0.0349 (0.0209)	0.0320 (0.0205)	0.0379 (0.0202)	0.0378 (0.0206)	0.0322 (0.0202)
ret_3	0.0069 (0.0410)	0.0077 (0.0419)	0.0030 (0.0422)	−0.0079 (0.0424)	0.0144 (0.0432)	−0.0119 (0.0420)
ret_6	0.0296 (0.0293)	0.0252 (0.0299)	0.0287 (0.0304)	0.0407 (0.0300)	0.0216 (0.0300)	0.0376 (0.0296)
Intercept	0.0168 (0.0336)	0.0204 (0.0339)	0.0174 (0.0347)	0.0036 (0.0340)	0.0244 (0.0340)	0.0108 (0.0342)
Observations	[N ₁ , N ₂ , N ₃ , N ₄ , N ₅ , N ₆]					

Notes: Each column represents a separate regression. Standard errors are in parentheses. Dashes indicate

that a variable was not included in that regression. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

Table 10 follows the same methodology as the previous table, but instead of aggregating signals using the mean, it sums the signals to account for the number of signals as well. This distinction is important because a mean value of 1 can result from either two investors or two hundred investors, yet the economic implications are different. By summing the signals, the measure reflects both the intensity and the breadth of investors' expectations about an industry. This approach differentiates between a scenario where only a few investors are optimistic versus a scenario where a large number of investors exhibit optimism, capturing the overall weight of market sentiment.

Table 10: Sum over change in tilts

Factor	(1)	(2)	(3)	(4)	(5)
RII_misinf	−0.0301 (0.0225)	— —	— —	— —	−0.0385 (0.0249)
RII_inf	— —	0.0400*** (0.0116)	— —	— —	0.0391*** (0.0126)
RII_inf_w	— —	— —	2.5370*** (0.7435)	— —	— —
RII_inf_w_	— —	— —	— —	0.4147*** (0.1258)	— —
Gat	−0.0468** (0.0195)	−0.0517*** (0.0193)	−0.0573*** (0.0190)	−0.0570*** (0.0194)	−0.0508** (0.0194)
divA_be	−0.3059** (0.1252)	−0.3076** (0.1291)	−0.2990** (0.1332)	−0.3115** (0.1342)	−0.2857** (0.1356)
me	-1.33×10^{-9} (9.01e-09)	7.91×10^{-10} (8.44e-09)	1.81×10^{-9} (8.53e-09)	1.18×10^{-9} (8.79e-09)	3.94×10^{-9} (9.52e-09)
profit	0.0463** (0.0201)	0.0416** (0.0211)	0.0375 (0.0214)	0.0407** (0.0220)	0.0387 (0.0224)
ret_3	0.0076 (0.0376)	0.0029 (0.0381)	−0.0003 (0.0391)	0.0023 (0.0401)	0.0005 (0.0389)
ret_6	0.0186 (0.0287)	0.0257 (0.0294)	0.0307 (0.0297)	0.0239 (0.0303)	0.0307 (0.0297)
Intercept	0.0265 (0.0329)	0.0204 (0.0338)	0.0158 (0.0341)	0.0226 (0.0347)	0.0137 (0.0341)
Observations	[N ₁ , N ₂ , N ₃ , N ₄ , N ₅]				

Notes: Each column represents a separate regression. Standard errors are in parentheses. Dashes indicate

that a variable was not included in that regression. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

Table 11 presents the results when the dependent variable is the raw industry return instead of excess returns. This specification examines whether the mean of changes in tilts, interpreted as signals, can predict future returns.

Table 11: Mean of Change in Tilts and Returns

$$R_{j,t+1} = \alpha + \gamma RII_{j,t} + \beta X_{j,t} + \epsilon_{j,t}$$

Factor	(1)	(2)	(3)	(4)	(5)
RII_misinf	−0.8831 (0.4539)	— —	−0.8286 (0.4958)	— —	— —
RII_inf	— —	1.2568*** (0.3517)	1.2570*** (0.3734)	— —	— —
RII_inf_w	— —	— —	— —	2.2071*** (0.6778)	— —
RII_inf_w_	— —	— —	— —	— —	0.4192*** (0.1266)
Intercept	0.0263 (0.0328)	0.0237 (0.0338)	0.0251 (0.0341)	0.0166 (0.0344)	0.0161 (0.0341)
divA_be	−0.3132** (0.1244)	−0.3045** (0.1305)	−0.2877** (0.1318)	−0.2978** (0.1344)	−0.3047** (0.1352)
me	-3.45×10^{-9} (8.84e-09)	7.46×10^{-10} (8.57e-09)	4.56×10^{-10} (9.20e-09)	1.71×10^{-9} (8.60e-09)	1.24×10^{-9} (8.87e-09)
profit	0.0466** (0.0199)	0.0419** (0.0215)	0.0386 (0.0219)	0.0374 (0.0217)	0.0412** (0.0222)
ret_3	0.0077 (0.0375)	0.0025 (0.0391)	0.0081 (0.0395)	−0.0014 (0.0396)	−0.0038 (0.0398)
ret_6	0.0190 (0.0285)	0.0225 (0.0296)	0.0210 (0.0298)	0.0297 (0.0299)	0.0291 (0.0300)
Gat	−0.0462** (0.0193)	−0.0509*** (0.0191)	−0.0513*** (0.0194)	−0.0556*** (0.0191)	−0.0523*** (0.0190)
Observations	[N ₁ , N ₂ , N ₃ , N ₄ , N ₅]				

Notes: Each column represents a separate regression. Standard errors are in parentheses. Dashes indicate

that a variable was not included in that regression. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

Table 12 presents the results when the filtering requirement is increased to 10 quarters. That is, a signal is considered informative only if it has been classified as such for at least 10 quarters.

Table 12: Changes-Filtered (10 Quarters) - Mean

round-mode = places, round-precision = 4,							
Factor	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RII_misinf	–	–	–	–1.3839*** (0.4127)	–1.3887*** (0.4398)	–1.5029*** (0.4363)	–1.4209*** (0.4582)
RII_inf	1.2373*** (0.2934)	–	–	–	1.4947*** (0.3414)	–	–
RII_inf_w	–	2.9327*** (0.6138)	–	–	–	2.8020*** (0.6420)	–
RII_inf_w_	–	–	0.3494** (0.1254)	–	–	–	0.4871*** (0.1651)
Intercept	0.0296 (0.0330)	0.0233 (0.0329)	0.0279 (0.0336)	0.0274 (0.0338)	0.0299 (0.0341)	0.0168 (0.0337)	0.0223 (0.0347)
divA_be	–0.3336*** (0.1187)	–0.3241** (0.1205)	–0.3343** (0.1210)	–0.3620*** (0.1311)	–0.3515** (0.1403)	–0.3363** (0.1434)	–0.3634*** (0.1451)
me	-3.19×10^{-9} (9.62e-09)	-5.50×10^{-9} (9.69e-09)	-2.92×10^{-9} (9.60e-09)	-1.64×10^{-9} (9.30e-09)	5.41×10^{-10} (1.09e-08)	-2.53×10^{-9} (1.08e-08)	1.35×10^{-10} (1.10e-08)
profit	0.0494** (0.0195)	0.0467** (0.0194)	0.0501** (0.0199)	0.0551*** (0.0208)	0.0496** (0.0231)	0.0482** (0.0232)	0.0561*** (0.0237)
ret_3	–0.0126 (0.0389)	–0.0158 (0.0382)	–0.0092 (0.0395)	0.0037 (0.0401)	–0.0016 (0.0430)	–0.0169 (0.0425)	–0.0086 (0.0443)
ret_6	0.0165 (0.0291)	0.0228 (0.0288)	0.0178 (0.0297)	0.0181 (0.0296)	0.0185 (0.0307)	0.0305 (0.0301)	0.0248 (0.0314)
Gat	–0.0429** (0.0187)	–0.0439** (0.0185)	–0.0454** (0.0189)	–0.0512*** (0.0193)	–0.0550*** (0.0211)	–0.0500*** (0.0207)	–0.0513*** (0.0214)
Observations	[N ₁ , N ₂ , N ₃ , N ₄ , N ₅ , N ₆ , N ₇]						

Notes: Each column represents a separate regression. Standard errors are in parentheses. Dashes indicate

that a variable was not included in that regression. Significance levels: *** p<0.01, ** p<0.05, * p<0.10.

Table 13 presents the regression results when the level of tilts, rather than changes in tilts, are used as signals. The findings indicate that informative signals do not predict returns, whereas uninformative and misinformative signals do.

Table 13: Regression Results when the Level of Tilts is the Signal

$$R_{j,t+1} = \alpha + \gamma RII_{j,t} + \beta X_{j,t} + \epsilon_{j,t}$$

Factor	(1)	(2)	(3)	(4)	(5)	(6)
RII_misinf	-0.0161*** (0.0055)	-0.0073 (0.0048)	— —	— —	-0.0135*** (0.0049)	— —
RII_inf	-0.0086 (0.0066)	— —	-0.0019 (0.0040)	— —	— —	-0.0079 (0.0048)
RII_zero	0.8786*** (0.3446)	— —	— —	— —	0.8785** (0.3446)	0.8406** (0.3393)
Intercept	0.0464*** (0.0113)	0.0488*** (0.0106)	0.0488*** (0.0107)	0.0486*** (0.0109)	0.0492*** (0.0110)	0.0470*** (0.0113)
divA_be	-0.2805** (0.1392)	-0.2829** (0.1309)	-0.2694** (0.1318)	-0.2764** (0.1336)	-0.2886** (0.1345)	-0.2938** (0.1389)
me	1.44e-09 (1.27e-08)	-2.00e-09 (9.93e-09)	-1.21e-09 (1.13e-08)	2.64e-09 (1.00e-08)	5.57e-09 (1.16e-08)	1.26e-08 (1.27e-08)
profit	0.0458** (0.0223)	0.0462** (0.0217)	0.0469** (0.0220)	0.0461** (0.0222)	0.0423* (0.0222)	0.0446** (0.0229)
Gat	-0.0433* (0.0226)	-0.0465** (0.0222)	-0.0444** (0.0218)	-0.0466** (0.0227)	-0.0529** (0.0220)	-0.0462** (0.0223)
Observations	—	—	—	—	—	—

Notes: This table reports the coefficients from cross-sectional regressions. Standard errors are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

3.1.4 Portfolio Analysis

In this section, I elaborate on the portfolio analysis. Five portfolios are formed by sorting on RII at each quarter. The first two columns in Table 14 report the portfolio returns and the number of industries in each portfolio. The portfolio of industries with the lowest demand shifts by their experts earns an average quarterly return of approximately 4.5 percent, whereas the portfolio of industries with the highest demand shifts by experts earns a quarterly return of 6 percent. The difference in returns between these two portfolios is about 1.5 percent per quarter.

The next columns present the regression results for the intercept (alpha) and the coefficients on six well-known risk factors. There are several key observations. First, all five portfolios exhibit a positive and significant alpha, suggesting that these portfolios generate excess returns beyond what is explained by traditional risk factors. Second, as the table indicates, portfolios formed based on industry demand shifts do not show meaningful covariation with the well-known risk factors, implying that these portfolios capture unique sources of variation in industry returns.

Table 14: Portfolio Returns, Industry Counts, and Regression Results

Portfolio	Returns	#Industries	Intercept	Mkt-RF	SMB	HML	RMW	CMA	WML
RII_1	4.55	9.58	4.23***	-0.05	-0.34	-0.03	0.25	-0.50	-0.07
RII_2	5.18	9.04	5.69***	-0.24	-0.03	0.03	-0.32	-0.70	-0.05
RII_3	5.47	9.01	4.96***	-0.05	-0.18	-0.34	0.44	-0.10	-0.12
RII_4	4.79	9.04	5.14***	-0.12	-0.27	-0.30	0.16	-0.39	-0.16
RII_5	6.02	9.52	6.12***	-0.10	-0.24	-0.17	0.09	-0.35	-0.16
5-1	1.46	-	1.13**	-0.04	0.15	-0.20	-0.13	0.20	-0.10

*** p<0.01, ** p<0.05, * p<0.1

Returns and industry counts are from 2002-07-01.

Table 15 presents the results when portfolios are sorted based on weighted signals. As expected, the gap in returns between the extreme portfolios increases significantly, highlighting the importance of weighting signals by their accuracy. The lowest signal portfolio earns an average quarterly return of approximately 4.03 percent, whereas the highest signal portfolio earns 6.20 percent. This 2.17 percent spread is notably larger than the unweighted signal portfolios.

The regression results on well-known risk factors show that these portfolios exhibit weak exposure to traditional risk factors. The intercept remains significant across all portfolios, suggesting that the signal captures unique predictive power beyond standard risk exposures.

Table 15: Portfolio Returns and Factor Regressions with Weighted Signals

Portfolio	Intercept	Mkt-RF	SMB	HML	RMW	CMA	WML
RII_1	4.03***	-0.06	-0.31	-0.18	0.31	-0.42	-0.09
RII_2	5.03***	-0.18	0.17	-0.40	0.12	-0.32	-0.11
RII_3	5.19***	-0.11	-0.09	-0.16	-0.15	-0.27	-0.15
RII_4	4.74***	0.10	-0.32	-0.43	0.63	-0.27	-0.10
RII_5	6.20***	-0.16	-0.24	-0.12	0.10	-0.43	-0.21
5-1	2.12*	-0.10	0.13	0.00	-0.17	0.04	-0.13

*** p<0.01, ** p<0.05, * p<0.1

Factor regressions include market (Mkt-RF), size (SMB), value (HML), profitability (RMW), investment (CMA), and momentum (WML).

Returns are estimated using weighted signals from 2002-07-01.

Table 16 presents the results when portfolios are sorted based on both criteria. The returns increase monotonically across the portfolios, with the highest signal portfolio earning a return of 5.63 percent per quarter, while the lowest signal portfolio earns 4.34 percent per quarter. This suggests that incorporating both dimensions enhances return predictability.

The regression results on standard risk factors show weak exposure to known risk factors, reinforcing the idea that these portfolios capture unique predictive power beyond traditional asset pricing models. The intercept remains positive and significant across all portfolios, indicating the persistence of return differentials.

Table 16: Factor Regressions for Portfolios Sorted on Both Criteria

Portfolio	Intercept	Mkt-RF	SMB	HML	RMW	CMA	WML
RII_1	4.34***	-0.05	-0.27	-0.13	0.20	-0.35	-0.12
RII_2	4.48***	-0.06	-0.30	0.02	0.23	-0.75**	-0.01
RII_3	5.32***	-0.14	-0.18	-0.23	0.05	-0.23	-0.20
RII_4	5.48***	0.01	-0.24	-0.42	0.18	-0.16	-0.06
RII_5	5.63***	-0.23	-0.37	-0.01	-0.01	-0.56	-0.07
Zero_inv	0.60	-0.17**	-0.05	0.06	-0.18	-0.15	0.04

*** p<0.01, ** p<0.05, * p<0.1

The table presents factor regressions for portfolios sorted on both dimensions.

Table 17: Factor Regressions for Portfolios Sorted on Both Criteria (Growth in Tilt as Signal)

Portfolio	Intercept	Mkt-RF	SMB	HML	RMW	CMA	WML
RIL_1	3.92***	-0.05	-0.22	-0.21	0.23	-0.38	-0.17
RIL_2	4.15***	0.02	-0.25	-0.09	0.20	-0.40	-0.06
RIL_3	5.19***	-0.07	-0.27	0.03	0.19	-0.44	-0.14
RIL_4	5.61***	-0.06	-0.46*	-0.04	0.07	-0.63*	-0.00
RIL_5	6.02***	-0.26*	-0.21	-0.12	-0.00	-0.35	-0.06
5-1	2.11**	-0.19**	0.05	0.02	-0.20	0.09	0.10

*** p<0.01, ** p<0.05, * p<0.1

The table presents factor regressions for portfolios sorted on both dimensions, where growth in tilt is used as the sorting signal.

3.2 Bayesian Methodology

Bayesian learning provides a probabilistic framework for updating beliefs about expected returns based on new information. The goal is to predict future industry returns, R_{t+1} , using investor expectations and past realized returns. I assume that the true industry return follows a distribution influenced by investor expectations, $E_t(R)$, and past realized returns, R_t :

$$R_{t+1} \sim P(R_{t+1}|\theta), \quad (13)$$

where θ represents the unknown parameters governing the relationship between returns, expectations, and past returns.

Before observing data, I assume a prior belief about the parameters θ :

$$P(\theta) \sim \mathcal{N}(\mu_0, \Sigma_0), \quad (14)$$

where μ_0 is the prior mean of the parameters, and Σ_0 is the prior covariance, capturing uncertainty in our initial belief.

Given investor expectations $E_t(R)$ and past returns R_t , we assume that returns follow a normal distribution:

$$R_{t+1}|E_t(R), R_t, \theta \sim \mathcal{N}(\beta_0 + \beta_1 E_t(R) + \beta_2 R_t, \sigma^2). \quad (15)$$

The corresponding likelihood function is:

$$P(R_{t+1}|E_t(R), R_t, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(R_{t+1} - (\beta_0 + \beta_1 E_t(R) + \beta_2 R_t))^2}{2\sigma^2}\right). \quad (16)$$

Using Bayes' theorem, I update our belief about θ given new observations:

$$P(\theta|R_{1:T}, E_{1:T}(R)) = \frac{P(R_{1:T}|\theta)P(\theta)}{P(R_{1:T})}. \quad (17)$$

If both the prior and likelihood are normal distributions, the posterior remains normal:

$$P(\theta|R_{1:T}, E_{1:T}(R)) \sim \mathcal{N}(\mu_T, \Sigma_T), \quad (18)$$

where

$$\mu_T = (\Sigma_0^{-1} + X^T X / \sigma^2)^{-1} (\Sigma_0^{-1} \mu_0 + X^T Y / \sigma^2), \quad (19)$$

$$\Sigma_T = (\Sigma_0^{-1} + X^T X / \sigma^2)^{-1}. \quad (20)$$

Here, $X = [1, E_t(R), R_t]$ is the matrix of explanatory variables, and $Y = R_{t+1}$ represents future realized returns.

The predictive distribution for R_{t+1} integrates over all possible parameter values:

$$P(R_{t+1}|R_{1:T}, E_{1:T}(R)) = \int P(R_{t+1}|\theta)P(\theta|R_{1:T}, E_{1:T}(R))d\theta. \quad (21)$$

Since the posterior distribution is normal, the predictive return follows:

$$R_{t+1}|R_{1:T}, E_{1:T}(R) \sim \mathcal{N}(\hat{\mu}_{T+1}, \hat{\sigma}_{T+1}^2), \quad (22)$$

where

$$\hat{\mu}_{T+1} = \beta_0 + \beta_1 E_t(R) + \beta_2 R_t, \quad (23)$$

$$\hat{\sigma}_{T+1}^2 = \sigma^2 + X_{T+1}^T \Sigma_T X_{T+1}. \quad (24)$$

This captures both model uncertainty and return volatility, unlike standard regression.

If investor expectations vary across industries, I can introduce industry-level priors:

$$\beta_i \sim \mathcal{N}(\mu_\beta, \tau^2), \quad (25)$$

where μ_β is the global mean effect of expectations across industries, and τ^2 captures industry-level heterogeneity. This allows each industry to have its own expected return dynamics.

If expectations evolve dynamically, we can model them using a Bayesian Kalman filter:

$$E_t(R) = \rho E_{t-1}(R) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_E^2). \quad (26)$$

This formulation dynamically adjusts the weight placed on investor expectations based on their accuracy over time.

Finally, if the relationship between expectations and returns is highly nonlinear, non-parametric Bayesian models such as Gaussian Processes (GPs) or Bayesian Neural Networks can be used to capture complex patterns.

This Bayesian learning framework provides a flexible and dynamic approach to forecasting industry returns. It incorporates investor expectations as a key predictive signal while continuously updating beliefs in response to new data. Unlike traditional regression approaches, the Bayesian method accounts for parameter uncertainty, making it particularly useful in financial forecasting where uncertainty is inherent.

3.3 Maximum Entropy

Maximum entropy provides a framework for predicting future returns by identifying the most unbiased probability distribution given known constraints. This approach is particularly useful when there is limited historical data, or when investor expectations and past returns contain uncertainty. The principle of maximum entropy states that, given a set of known constraints, the probability distribution that best represents our knowledge is the one that maximizes entropy.

The entropy of a probability distribution $P(R)$ for future returns R is defined as:

$$H(P) = - \sum_i P(R_i) \log P(R_i). \quad (27)$$

The goal is to estimate $P(R)$ subject to known constraints, ensuring that the chosen distribution remains as uninformative as possible while still satisfying the given information. The constraints typically include the expected return and variance:

$$\mathbb{E}[R] = \sum_i R_i P(R_i), \quad (28)$$

$$\text{Var}(R) = \sum_i (R_i - \mathbb{E}[R])^2 P(R_i). \quad (29)$$

To find the maximum entropy distribution, we solve the constrained optimization problem:

$$\max_{P(R)} - \sum_i P(R_i) \log P(R_i), \quad (30)$$

subject to:

$$\sum_i P(R_i) = 1, \quad (31)$$

$$\sum_i R_i P(R_i) = \mathbb{E}[R], \quad (32)$$

$$\sum_i (R_i - \mathbb{E}[R])^2 P(R_i) = \sigma^2. \quad (33)$$

Using the method of Lagrange multipliers, we define the Lagrangian:

$$\begin{aligned} \mathcal{L} = & - \sum_i P(R_i) \log P(R_i) + \lambda_0 \left(1 - \sum_i P(R_i) \right) \\ & + \lambda_1 \left(\mathbb{E}[R] - \sum_i R_i P(R_i) \right) + \lambda_2 \left(\sigma^2 - \sum_i (R_i - \mathbb{E}[R])^2 P(R_i) \right). \end{aligned} \quad (34)$$

Differentiating with respect to $P(R_i)$ and solving for the probability distribution, we obtain:

$$P(R_i) = \frac{1}{Z} \exp(\lambda_1 R_i + \lambda_2 R_i^2), \quad (35)$$

where Z is the normalization constant ensuring that probabilities sum to one:

$$Z = \sum_i \exp(\lambda_1 R_i + \lambda_2 R_i^2). \quad (36)$$

The parameters λ_1 and λ_2 are estimated by solving the moment constraints for expected return and variance.

To predict future returns, we compute the expectation under the estimated maximum entropy distribution:

$$R_{t+1} = \sum_i R_i P(R_i | R_{1:T}, E_{1:T}(R)). \quad (37)$$

This approach ensures that the predicted return distribution reflects all available infor-

mation while making minimal additional assumptions. Unlike traditional regression models, maximum entropy does not impose a parametric structure on returns, making it a flexible tool for financial forecasting.

When investor expectations are incorporated, we extend the constraint set to include moments of the expectation distribution:

$$\sum_i E_t(R_i)P(R_i) = \mathbb{E}[E_t(R)]. \quad (38)$$

This formulation allows the model to balance between historical returns and investor expectations, adjusting return predictions based on the degree of disagreement in expectations.

A key advantage of maximum entropy is its ability to handle cases where data is sparse or where the return-generating process is complex. The method can be further extended to dynamic settings by incorporating time-varying constraints, such as a Bayesian updating mechanism that adjusts the return distribution as new information becomes available.

In settings where investor expectations are highly dispersed, entropy serves as a measure of information uncertainty, capturing the degree of disagreement among market participants. Higher entropy in investor expectations may indicate greater mispricing, suggesting that future returns may deviate from naive historical projections.

This framework provides a nonparametric alternative to traditional econometric models, ensuring that return predictions remain unbiased while fully utilizing the information embedded in both past returns and investor expectations.

4 Conclusion

This paper examines the informational role of institutional investors' demand for industries.

More specifically, I test whether "who owns what", is actually incorporated into prices. To do this, I interpret quarterly changes in investors' demand as signals about future industry prices. To assess the accuracy of these signals, I introduce a method that evaluates individual investors' predictive performance and aggregates their signals at the industry level. The resulting measure, Revealed Industry Information (RII), serves as a proxy for expected industry returns, capturing price-relevant insights embedded in institutional demand.

Empirical results show that this quantified information robustly predicts industry returns, revealing industry return patterns that standard models fail to capture and suggesting that investors' holdings reveal information orthogonal to well-known information sets used in standard models

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